

Residual Human–Robot Interaction Modeling

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THESIS CERTIFICATE

This is to certify that the thesis titled **Residual Human–Robot Interaction Modeling**, submitted by **Nishank Shakyawar (2024JRB2032)**, to the Indian Institute of Technology, Delhi, for the award of the degree of **Master of Technology**, is a bonafide record of the work done by him under our supervision at the **CoE BIRD**. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Musculoskeletal simulation has become a key tool in biomechanics, rehabilitation engineering, wearable robotics, and human movement analysis. Platforms like OpenSim offer detailed estimates of muscle activations, joint torques, and other biomechanical quantities that are otherwise hard to measure directly. Yet simulation outputs often don't match experimentally observed physiological signals, largely because of model simplifications, individual variability, measurement noise, and biological processes that aren't fully captured by the model. These gaps limit how confidently simulation results can be used in safety-critical settings such as assistive robotics and rehabilitation systems.

This dissertation introduces a new uncertainty-aware framework for musculoskeletal modelling that brings together biomechanical simulation, experimental electromyography (EMG), deep learning, residual modelling, and uncertainty quantification. The framework combines motion capture data from the Vicon Nexus system with EMG recordings collected during upper-limb biceps curl exercises. Kinematic data from these experiments are processed in OpenSim to generate joint torques and muscle activations via inverse kinematics, inverse dynamics, and static optimization.

To close the gap between simulated and physiological muscle behaviour, the study explores several deep learning architectures — Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Transformers, and hybrid CNN-LSTM models. A residual learning approach is used to capture the discrepancies between simulated and experimentally measured muscle activations. The framework also incorporates aleatoric and epistemic uncertainty estimation, allowing it to quantify prediction confidence and improve interpretability.

The experimental results show strong predictive performance across several muscle groups, with coefficients of determination exceeding 0.84 and reaching close to 0.93 for some channels. Residual learning notably improves prediction accuracy, while decomposing uncertainty into its aleatoric and epistemic components yields meaningful confidence estimates that distinguish between measurement noise and model-related uncertainty.

Overall, this methodology lays a solid foundation for future work in wearable robotics, rehabilitation engineering, human–robot interaction, and real-time digital human twins. By combining physics-based biomechanical modelling with uncertainty-aware AI, the framework contributes to the development of more trustworthy and interpretable systems for predicting human movement.

Keywords: Musculoskeletal Modelling, OpenSim, Electromyography, Deep Learning, CNN-LSTM, Uncertainty Quantification, Human–Robot Interaction, Wearable Robotics, Digital Twin, Residual Learning.

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ABBREVIATIONS

AI	Artificial Intelligence
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
CI	Confidence Interval
DL	Deep Learning
DOF	Degree of Freedom
EDA	Exploratory Data Analysis
EEG	Electroencephalography
EMG	Electromyography
FFT	Fast Fourier Transform
GPU	Graphics Processing Unit
HRI	Human-Robot Interaction
IK	Inverse Kinematics
IMU	Inertial Measurement Unit
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MC	Monte Carlo
MVC	Maximum Voluntary Contraction
MSE	Mean Squared Error
PSD	Power Spectral Density
PYTORCH	Python Torch Framework
R²	Coefficient of Determination
RELU	Rectified Linear Unit
RMSE	Root Mean Square Error
RMS	Root Mean Square
SO	Static Optimization
UQ	Uncertainty Quantification

NOTATION

x	Input Feature Vector
y	Actual Output
$\hat{y}$	Predicted Output
r_{EMG}	Residual EMG Error
ϑ	Neural Network Parameters
σ_a^2	Aleatoric Variance
σ_e^2	Epistemic Variance
τ	Joint Torque
ω	Angular Velocity
α	Angular Acceleration
I	Moment of Inertia
PSD	Power Spectral Density
$RMSE$	Root Mean Square Error
MAE	Mean Absolute Error
R^2	Coefficient of Determination
r	Pearson Correlation Coefficient
CI	Confidence Interval
L	Loss Function
T	Number of Monte Carlo Samples
N	Number of Data Samples
$f(x)$	Prediction Function
$\text{Var}(y x)$	Predictive Variance of Output Given Input

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Recent developments in wearable robotics have significantly advanced the fields of rehabilitation engineering, human augmentation, and assistive technologies. Devices such as robotic exoskeletons and soft exosuits are increasingly being designed to assist individuals with motor impairments, reduce physical fatigue in industrial environments, and improve human performance during repetitive or physically demanding tasks. These systems provide mechanical assistance while allowing users to perform movements that closely resemble their natural motion.

A major challenge in designing such systems lies in accurately understanding the interaction between the human musculoskeletal system and the wearable device. Human movement is governed by a complex coordination of muscles, tendons, joints, and the nervous system. Since this interaction varies across individuals and is influenced by several physiological and environmental factors, developing accurate computational models remains a difficult task.

Physics-based musculoskeletal simulation platforms, particularly OpenSim, have become an important tool for studying human movement. OpenSim enables researchers to estimate joint torques, muscle activations, muscle forces, tendon dynamics, and metabolic energy expenditure using anatomically detailed musculoskeletal models. These capabilities have made it widely applicable in biomechanics, rehabilitation research, sports science, and human–robot interaction.

Although these simulation models provide accurate biomechanical estimates, they are computationally intensive and depend heavily on accurate model parameters and high-quality input measurements. Small errors in experimental data or model scaling may lead to noticeable variations in simulation results. Moreover, the computational time required for inverse dynamics, static optimization, and forward simulations makes conventional approaches unsuitable for applications that require real-time decision making.

The rapid progress in deep learning provides an attractive alternative for addressing these limitations. Neural networks are capable of learning complex nonlinear relationships directly from data and can generate predictions within milliseconds once trained. By learning the mapping between experimentally measured signals and OpenSim-generated biomechanical outputs, these models can serve as fast surrogate models without repeatedly performing computationally expensive simulations.

Motivated by this need, the present work investigates the use of deep learning techniques to predict muscle activity, approximate OpenSim outputs, and quantify the uncertainty associated with these predictions. The study further examines the influence of robotic assistance on muscle behavior by integrating motion capture data, electromyography (EMG) measurements, and biomechanical

simulations for upper-limb biceps curl movements. The long-term objective is to establish a reliable framework that supports accurate and uncertainty-aware control strategies for wearable robotic systems operating in real time.

1.2 Problem Statement

Accurate estimation of muscle activity during human movement remains an open challenge in biomechanics and wearable robotics. While OpenSim provides detailed estimates of muscle activations and joint dynamics, its computational requirements make it difficult to deploy in applications where rapid response is essential. Furthermore, simulated muscle activations often differ from experimentally recorded EMG signals because of physiological variability, sensor noise, simplifications in musculoskeletal models, parameter estimation errors, and individual adaptation during movement.

The introduction of wearable robotic devices adds another layer of complexity. Factors such as exosuit misalignment, controller latency, mechanical compliance, interaction forces, and changes in user behavior introduce additional uncertainty into the estimation process. As a result, assistive robotic systems frequently operate without complete knowledge of the actual physiological state of the user.

This research addresses these challenges through two complementary objectives. The first is to develop deep learning models capable of predicting muscle behavior directly from biomechanical measurements while closely approximating the outputs generated by OpenSim. The second is to model the discrepancy between simulated and experimentally measured muscle activity by explicitly quantifying prediction uncertainty.

The residual muscle activity is defined as

$$r_{EMG} = |EMG_{actual} - EMG_{simulated}|$$

where EMG_{actual} represents the experimentally recorded muscle activity, $EMG_{simulated}$ denotes the corresponding OpenSim prediction, and r_{EMG} represents the residual error. Modeling this residual enables the proposed framework to generate more reliable muscle predictions suitable for real-time wearable robotic applications.

1.3 Research Objectives

The primary objective of this research is to develop an uncertainty-aware framework for musculoskeletal modeling that combines biomechanical simulation with modern deep learning techniques for upper-limb movements.

To achieve this goal, the study first develops a comprehensive dataset containing shoulder and elbow joint angles, joint torques, simulated EMG signals, experimentally recorded EMG data, and external

load information collected during biceps curl exercises. Using these measurements, an OpenSim-based simulation pipeline is established to estimate joint torques, muscle activations, muscle forces, and metabolic cost.

The study further investigates multiple deep learning architectures, including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Transformer models, hybrid CNN–LSTM and CNN–Transformer architectures, and residual neural networks for predicting muscle behavior. Special emphasis is placed on residual learning techniques that model the differences between simulated and measured muscle activity.

Another major objective is to quantify prediction uncertainty by separating the total predictive variance into aleatoric and epistemic components. This decomposition is expressed as

$$Var(y | x) = \sigma_a^2(x) + \sigma_e^2(x)$$

where σ_a^2 represents aleatoric uncertainty arising from measurement noise and inherent variability, while σ_e^2 represents epistemic uncertainty associated with the learning model. Finally, the study analyzes the influence of robotic assistance on human movement and establishes a foundation for future real-time exosuit control systems.

CHAPTER 2: RELATED WORK

2.1 Introduction

Research on human movement has progressed considerably over the past few decades owing to advances in biomechanics, wearable sensing technologies, robotics, and artificial intelligence. These developments have enabled researchers to investigate human motion with greater precision and have significantly influenced the design of rehabilitation systems, assistive devices, and human performance enhancement technologies. Modern biomechanical studies commonly integrate motion capture systems, electromyography (EMG), musculoskeletal simulation software, and machine learning techniques to better understand the complex interaction between muscles, joints, and external mechanical assistance.

The growing adoption of wearable robotic systems, including rigid exoskeletons and soft exosuits, has further increased the demand for accurate estimation of muscle activations, joint forces, interaction loads, and other biomechanical variables. Reliable estimation of these quantities is essential for developing intelligent control strategies that can provide effective assistance while preserving natural human movement. Traditionally, physics-based musculoskeletal models have been used for this purpose because they provide physiologically meaningful estimates of internal body dynamics. However, these approaches often require substantial computational resources and are difficult to deploy in real-time applications.

Recent advances in artificial intelligence have introduced data-driven alternatives capable of learning complex nonlinear relationships directly from experimental data. Deep learning models have demonstrated promising performance in predicting biomechanical variables while significantly reducing computational time. Nevertheless, both physics-based and learning-based approaches remain affected by uncertainties arising from measurement noise, subject-specific variability, simplifications in musculoskeletal models, and environmental factors. As wearable robotic systems increasingly interact with humans in real-world environments, understanding and quantifying these uncertainties has become an important research objective.

2.2 Historical Development

2.2.1 Early Biomechanical Modeling

The foundations of modern biomechanical analysis can be traced back to the 1960s and 1970s, when researchers began applying principles of rigid-body mechanics to study human movement. Early biomechanical models represented the human body as a system of interconnected rigid segments linked by joints. These models enabled the estimation of joint kinematics, segmental motion, ground reaction forces, and overall gait characteristics through inverse dynamics analysis. Although these methods provided valuable insights into external body motion, they were unable to estimate the internal forces

generated by muscles and tendons, thereby limiting their applicability in rehabilitation and movement analysis.

2.2.2 Emergence of Musculoskeletal Models

During the 1980s and 1990s, biomechanical research shifted toward anatomically realistic musculoskeletal models capable of representing muscle–tendon dynamics. These models incorporated Hill-type muscle formulations, tendon elasticity, and neuromuscular activation mechanisms, enabling researchers to estimate muscle forces, muscle activations, joint reaction forces, and metabolic energy expenditure. This transition marked a significant advancement because it allowed investigations to move beyond external motion analysis and explore the internal physiological mechanisms responsible for human movement.

Among the major milestones during this period was the work of Scott Delp and his collaborators, whose research ultimately led to the development of OpenSim. This platform has since become one of the most widely adopted software environments for musculoskeletal simulation and remains an essential tool in biomechanics and rehabilitation research.

2.2.3 Development of Motion Capture Systems

The introduction of optical motion capture systems transformed experimental biomechanics by enabling highly accurate measurement of human movement. Commercial systems such as Vicon, Qualisys, and OptiTrack provide high-resolution tracking of reflective markers attached to anatomical landmarks. These systems allow researchers to reconstruct joint trajectories, estimate angular velocities and accelerations, and generate precise kinematic data required for biomechanical simulations.

Motion capture technology has become an integral component of musculoskeletal modeling because accurate kinematic measurements directly influence the reliability of inverse dynamics and muscle force estimation. In the present research, upper-limb movement data were collected using the Vicon Nexus motion capture system, which provided synchronized measurements for subsequent OpenSim simulations.

2.2.4 Evolution of Electromyography

Electromyography has undergone substantial technological development, progressing from simple analog recording systems to sophisticated multi-channel digital acquisition platforms. Modern EMG systems enable non-invasive monitoring of muscle activation with high temporal resolution, making them valuable for investigating neuromuscular function during human movement.

Over the years, EMG has found widespread applications in rehabilitation engineering, prosthetic control, human–machine interfaces, fatigue assessment, and wearable robotic control. In upper-limb biomechanics, EMG provides direct information regarding muscle recruitment patterns that cannot be

obtained solely from kinematic measurements. The present study utilizes EMG recordings from six upper-limb muscles—Biceps Long Head, Biceps Short Head, Brachialis, Triceps Long Head, Triceps Lateral Head, and Triceps Medial Head—to analyze muscle behavior during biceps curl exercises performed under both human-only and exosuit-assisted conditions.

2.2.5 Emergence of Wearable Robotics

The field of wearable robotics experienced rapid expansion after the early 2000s with the development of powered exoskeletons and lightweight soft exosuits. Initially designed for military applications, these systems were intended to reduce the physical burden associated with carrying heavy loads and performing repetitive tasks. Over time, their applications expanded into healthcare, rehabilitation, and industrial environments.

In rehabilitation, wearable robotic devices assist patients recovering from stroke, spinal cord injury, Parkinson's disease, and other neuromuscular disorders by providing controlled mechanical support during therapeutic exercises. Industrial exoskeletons aim to reduce worker fatigue, improve ergonomic safety, and minimize the risk of musculoskeletal injuries during repetitive lifting tasks. Similar technologies have also been investigated for military personnel to improve endurance, reduce fatigue, and enhance operational efficiency.

The increasing adoption of wearable robotic systems has created new research challenges related to accurately modeling human–robot interaction, estimating muscle activity in real time, and designing adaptive control strategies capable of responding to individual user characteristics.

2.3 Literature Review

2.3.1 OpenSim-Based Musculoskeletal Modeling

OpenSim has become one of the most widely used open-source platforms for musculoskeletal simulation because of its flexibility, extensibility, and physiologically realistic modeling capabilities. The software provides a comprehensive framework for analyzing human movement by integrating anatomical models with inverse kinematics, inverse dynamics, static optimization, and forward dynamic simulations.

Delp et al. (2007) introduced OpenSim as a general-purpose simulation environment capable of representing complex musculoskeletal systems using customizable anatomical models. Its open-source architecture enabled researchers to modify model parameters, incorporate new muscles and joints, and develop application-specific simulation pipelines. As a result, OpenSim rapidly became a standard platform for research in rehabilitation engineering, sports biomechanics, orthopedic analysis, and human–robot interaction.

Building upon this foundation, Seth et al. (2018) significantly enhanced OpenSim by improving computational performance, introducing more efficient optimization algorithms, and expanding scripting capabilities through programming interfaces. These developments simplified automation of large simulation studies while improving computational efficiency for musculoskeletal analysis.

Today, OpenSim remains one of the most extensively validated platforms for estimating muscle activations, joint torques, muscle forces, tendon dynamics, and metabolic energy expenditure. Despite its accuracy, the computational cost associated with repeated optimization procedures continues to limit its application in real-time wearable robotic systems.

2.3.2 Real-Time Musculoskeletal Simulation

Achieving real-time musculoskeletal simulation has become an important objective in wearable robotics because assistive devices require rapid estimation of biomechanical states for effective control. Traditional OpenSim workflows involve multiple computationally demanding optimization steps, making them unsuitable for applications requiring immediate feedback.

Pizzolato et al. (2017) addressed this challenge by developing a real-time OpenSim framework capable of performing inverse kinematics and inverse dynamics with latencies below 31.5 milliseconds while processing approximately 2000 frames per second. Their work demonstrated that online biomechanical analysis could be achieved under simplified modeling conditions, thereby opening new opportunities for rehabilitation feedback systems and wearable robotic control.

However, the authors also reported that computational performance declined considerably when more complex muscle models or optimization routines were incorporated. Consequently, achieving high physiological accuracy while maintaining real-time computational performance remains an open research challenge.

2.3.3 Marker- and IMU-Based Biomechanical Systems

Although laboratory-based motion capture systems provide highly accurate measurements, their dependence on specialized equipment limits their use in everyday environments. To overcome this limitation, researchers have explored the integration of inertial measurement units (IMUs) with optical motion capture systems for portable biomechanical analysis.

Stanev et al. (2021) proposed a framework combining IMUs and optical markers to estimate joint moments, muscle forces, and other biomechanical variables outside conventional laboratory settings. Their work demonstrated the feasibility of wearable rehabilitation systems capable of providing continuous biomechanical feedback during daily activities. Nevertheless, challenges such as sensor drift, calibration errors, and inaccuracies in estimating ground reaction forces continue to affect the reliability of portable motion analysis systems.

2.3.4 Virtual Muscle Force Sensing

Virtual sensing has recently emerged as an alternative approach for estimating muscle forces without relying exclusively on direct physiological measurements. Instead of measuring internal muscle forces experimentally, virtual sensing combines mathematical models with sensor data to infer muscle behavior.

Adduci and Mundo (2024) proposed a virtual muscle force sensing framework for upper-limb rehabilitation that integrates multibody dynamics with Kalman filtering techniques. Their approach achieved root mean square errors close to 1.5% for selected muscles, demonstrating the potential of model-based estimation methods in rehabilitation applications. Despite these encouraging results, the method remains sensitive to sensor placement and suffers from observability issues when limited measurements are available. These limitations highlight the need for more robust prediction frameworks capable of handling measurement uncertainty while maintaining high computational efficiency.

2.4 EMG-Based Human Movement Modeling

Electromyography (EMG) has become one of the most widely used techniques for studying neuromuscular activity because it provides a direct measure of muscle activation during voluntary movement. Unlike motion capture systems that primarily record external kinematics, EMG reflects the electrical activity generated within muscles, allowing researchers to investigate motor control strategies, muscle coordination, fatigue, and neuromuscular disorders. Consequently, EMG has found extensive applications in biomechanics, rehabilitation engineering, prosthetic control, sports science, and wearable robotics.

In recent years, researchers have increasingly combined EMG with musculoskeletal simulation to obtain a more comprehensive understanding of human movement. While simulation platforms such as OpenSim estimate muscle activations based on biomechanical principles, experimentally recorded EMG serves as an important physiological reference for validating simulation results. Comparing simulated and measured muscle activity enables researchers to identify modeling inaccuracies and improve the reliability of musculoskeletal predictions.

Despite its usefulness, EMG signals are inherently noisy and are influenced by several physiological and environmental factors. Motion artifacts, power-line interference, electrode displacement, variations in skin impedance, muscle cross-talk, and changes in muscle fatigue can all affect signal quality. Consequently, appropriate preprocessing is essential before EMG data can be used for biomechanical analysis or machine learning applications.

A standard preprocessing pipeline generally begins with notch filtering to remove power-line interference. Depending on the electrical supply frequency, filters operating at either 50 Hz or 60 Hz are commonly employed. Since the experiments conducted in this research were performed under a 50

Hz power supply, a 50 Hz notch filter was applied to eliminate electrical noise without affecting the underlying muscle activity.

Following notch filtering, band-pass filtering is performed to retain the frequency components associated with muscle activation while suppressing low-frequency movement artifacts and high-frequency electronic noise. Most EMG studies employ a frequency range between 20 Hz and 450 Hz because it effectively captures the dominant frequency content of surface EMG signals.

The filtered EMG signal is then transformed into a smooth representation of muscle activation through rectification and Root Mean Square (RMS) envelope extraction. RMS processing reduces high-frequency fluctuations while preserving the overall activation pattern, making the signal more suitable for biomechanical interpretation and neural network training. In the present work, RMS envelopes were computed using a moving window of 50 ms, providing a balance between temporal resolution and signal stability.

The integration of properly processed EMG signals with biomechanical simulations has become an important research direction because it combines physiological measurements with physics-based models. This combination enables the development of more accurate predictive models while also supporting the analysis of discrepancies between simulated and experimentally measured muscle activity, which forms a central objective of the present research.

2.5 Deep Learning in Biomechanics

The increasing availability of large biomechanical datasets has encouraged researchers to adopt deep learning techniques for modeling complex relationships between physiological signals, joint kinematics, muscle activations, and external forces. Unlike traditional regression methods that rely on handcrafted features, deep neural networks automatically learn hierarchical representations from data, allowing them to capture highly nonlinear biomechanical behavior. As a result, deep learning has become an important tool for predicting muscle activity, estimating joint torques, recognizing movement patterns, and developing intelligent control strategies for wearable robotic systems.

Various neural network architectures have been investigated for biomechanical applications, each offering distinct advantages depending on the characteristics of the input data and the prediction task.

2.5.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks were originally developed for computer vision applications but have since been successfully adapted for one-dimensional physiological signals such as EMG and kinematic time series. CNNs employ convolutional filters that automatically extract meaningful local features from the input signal without requiring manual feature engineering. This capability makes them particularly effective for identifying characteristic activation patterns within muscle signals.

Several studies have demonstrated that CNN-based models achieve high prediction accuracy while maintaining relatively low computational complexity. Their shared-weight architecture significantly reduces the number of trainable parameters compared with fully connected networks, enabling efficient learning even from moderately sized datasets. Furthermore, convolutional operations provide a degree of robustness to measurement noise, which is particularly beneficial when working with physiological signals.

Despite these advantages, CNNs primarily focus on local spatial or temporal patterns and have limited ability to capture long-range temporal dependencies. Since human movement is inherently sequential, this limitation often restricts their performance when long-duration motion patterns must be modeled.

2.5.2 Long Short-Term Memory Networks (LSTMs)

Long Short-Term Memory (LSTM) networks were introduced to overcome the limitations of conventional recurrent neural networks in modeling sequential data. Through specialized memory cells and gating mechanisms, LSTMs are capable of retaining information over extended time intervals while selectively forgetting irrelevant information. This property makes them particularly suitable for analyzing continuous biomechanical signals.

LSTM networks have been widely applied in gait prediction, EMG forecasting, joint angle estimation, and force prediction because they effectively capture temporal relationships between consecutive movement samples. Their ability to model long-range dependencies enables more accurate prediction of muscle behavior during dynamic activities compared with purely convolutional approaches.

However, LSTM models generally require longer training times and higher computational resources than CNNs. In addition, training recurrent networks on long sequences may become computationally expensive, especially when large datasets are used. These challenges have motivated researchers to explore alternative architectures capable of maintaining temporal modeling performance while improving computational efficiency.

2.5.3 Transformer Networks

Transformer architectures represent one of the most significant recent developments in deep learning. Unlike recurrent networks, Transformers rely on self-attention mechanisms to model relationships among all elements within an input sequence simultaneously. This design allows them to capture both local and global dependencies while enabling efficient parallel computation during training.

In biomechanics, Transformer models have recently been applied to motion prediction, activity recognition, sensor fusion, and sequential physiological signal analysis. Their ability to learn complex interactions among multiple input variables makes them particularly attractive for modeling human movement, where numerous biomechanical variables interact simultaneously.

Despite these advantages, Transformer models typically require substantially larger training datasets than CNNs or LSTMs. Their computational complexity also increases with sequence length, making them more demanding in terms of memory and processing resources. Consequently, their effectiveness depends on the availability of sufficient training data and appropriate computational infrastructure.

2.5.4 Hybrid CNN–LSTM Architectures

To exploit the complementary strengths of convolutional and recurrent neural networks, recent research has increasingly focused on hybrid CNN–LSTM architectures. In these models, convolutional layers first extract informative local features from biomechanical signals, while LSTM layers subsequently learn the temporal relationships among those extracted features. This hierarchical processing enables the network to model both short-term signal characteristics and long-term movement dynamics.

Several studies have reported that hybrid CNN–LSTM models outperform standalone CNN or LSTM networks in applications involving EMG classification, gait analysis, force estimation, and movement prediction. The convolutional component effectively suppresses noise and identifies meaningful activation patterns, whereas the recurrent component captures sequential dependencies that are essential for representing continuous human motion.

Owing to these advantages, hybrid CNN–LSTM architectures have become one of the most promising approaches for biomechanical prediction tasks. The present research adopts this architecture as one of the primary models for predicting muscle activity and estimating uncertainty during upper-limb movements. By combining efficient feature extraction with temporal sequence modeling, the proposed framework seeks to improve prediction accuracy while maintaining computational efficiency suitable for future real-time wearable robotic systems.

2.6 Uncertainty Quantification

The increasing adoption of artificial intelligence in healthcare, biomechanics, and wearable robotics has highlighted the importance of uncertainty quantification (UQ). While deep learning models have demonstrated remarkable predictive capabilities, they often provide deterministic outputs without indicating the confidence associated with those predictions. In safety-critical applications such as rehabilitation robotics and human–robot interaction, inaccurate predictions can adversely affect both system performance and user safety. Consequently, recent research has focused not only on improving prediction accuracy but also on estimating the uncertainty associated with model predictions.

In biomechanical applications, uncertainty arises from multiple sources, including measurement errors, physiological variability, model simplifications, and limited training data. Quantifying these uncertainties allows researchers to evaluate the reliability of predicted muscle activations, joint forces, and interaction torques, thereby enabling more informed decision-making in assistive robotic systems.

Prediction uncertainty is generally categorized into two major components: **aleatoric uncertainty** and **epistemic uncertainty**. Although both contribute to the total prediction variance, they originate from different sources and therefore require different modeling approaches.

2.6.1 Aleatoric Uncertainty

Aleatoric uncertainty represents the inherent randomness present in experimental observations and measurement processes. Since this uncertainty originates from the data itself, it cannot be completely eliminated even if additional training data become available. In biomechanics, aleatoric uncertainty is primarily associated with the natural variability of human movement and imperfections in sensing systems.

Surface EMG recordings provide a clear example of this type of uncertainty. Factors such as electrode displacement, muscle cross-talk, changes in skin impedance, perspiration, motion artifacts, and physiological fatigue can introduce considerable variability into EMG measurements. Similarly, motion capture systems may experience marker occlusion, calibration inaccuracies, or soft tissue artifacts that influence the accuracy of recorded joint kinematics.

Because these sources of uncertainty are unavoidable during experimental data collection, they are commonly regarded as irreducible. Instead of attempting to eliminate them, modern machine learning approaches explicitly model aleatoric uncertainty to improve the robustness of biomechanical predictions. Estimating this uncertainty enables predictive models to distinguish between reliable observations and noisy measurements, resulting in more stable performance during practical deployment.

2.6.2 Epistemic Uncertainty

Unlike aleatoric uncertainty, epistemic uncertainty originates from the predictive model itself. It reflects incomplete knowledge of the underlying relationship between input variables and desired outputs. This type of uncertainty typically arises because of limited training data, insufficient model complexity, poor parameter estimation, or unseen operating conditions.

In wearable robotics, epistemic uncertainty becomes particularly significant when prediction models encounter movement patterns that differ from those observed during training. Variations in subject anatomy, muscle recruitment strategies, external loading conditions, or robotic assistance levels may all lead to reduced model confidence. Unlike measurement noise, epistemic uncertainty can be reduced by collecting additional training data, improving network architectures, or employing more effective optimization strategies.

Estimating epistemic uncertainty provides valuable information regarding the reliability of model predictions. Rather than producing a single deterministic estimate, uncertainty-aware models can

identify situations where predictions should be interpreted cautiously or where additional sensor information may be required before control decisions are made.

2.6.3 Bayesian Deep Learning

Bayesian Deep Learning has emerged as one of the most widely investigated approaches for uncertainty estimation in neural networks. Unlike conventional deep learning models, which estimate fixed parameter values during training, Bayesian methods represent model parameters as probability distributions. This probabilistic formulation enables uncertainty to be propagated throughout the prediction process, resulting in confidence estimates alongside predicted outputs.

Several Bayesian approaches have been proposed for biomechanical and medical applications. Bayesian Neural Networks explicitly learn probability distributions over network weights, while Variational Inference approximates these distributions through optimization techniques that reduce computational complexity. Deep Ensemble methods estimate uncertainty by combining predictions from multiple independently trained models, thereby capturing variability arising from different parameter initializations.

Although Bayesian methods provide theoretically rigorous uncertainty estimates, they often involve substantial computational overhead, making their application challenging for large-scale neural networks and real-time wearable robotic systems. Consequently, researchers have investigated alternative techniques that provide reliable uncertainty estimates while remaining computationally efficient.

2.6.4 Monte Carlo Dropout

Monte Carlo (MC) Dropout has become one of the most practical methods for estimating epistemic uncertainty in deep learning. Originally introduced as a regularization technique to reduce overfitting, dropout randomly deactivates neurons during training. Gal and Ghahramani later demonstrated that applying dropout during both training and inference approximates Bayesian inference in deep neural networks.

Instead of generating a single prediction during inference, MC Dropout performs multiple forward passes through the network using different dropout masks. The distribution of these predictions provides an estimate of model uncertainty, where the mean represents the predicted value and the variance reflects epistemic uncertainty.

Compared with Bayesian Neural Networks, MC Dropout is considerably easier to implement because it requires only minor modifications to standard neural network architectures. It has therefore been successfully applied to medical image analysis, autonomous driving, robotics, and increasingly to biomechanics and rehabilitation engineering.

In the present research, MC Dropout is employed to estimate epistemic uncertainty associated with muscle activity prediction. The resulting confidence estimates provide additional information regarding prediction reliability, enabling more informed interpretation of biomechanical outputs.

2.6.5 Uncertainty Decomposition

Recent research has increasingly emphasized the importance of separating total prediction uncertainty into its aleatoric and epistemic components. Such decomposition allows researchers to distinguish between uncertainty arising from measurement noise and uncertainty resulting from limitations of the predictive model.

The overall predictive variance can be expressed as

$$Var(y | x) = \sigma_a^2(x) + \sigma_e^2(x)$$

where $\sigma_a^2(x)$ denotes aleatoric uncertainty and $\sigma_e^2(x)$ represents epistemic uncertainty.

This decomposition is particularly valuable in wearable robotic applications because it enables engineers to identify the dominant source of prediction error. If uncertainty is primarily aleatoric, improvements in sensing hardware or experimental protocols may be required. Conversely, if epistemic uncertainty dominates, additional training data or more expressive neural network architectures may significantly improve predictive performance.

The ability to distinguish between these two uncertainty sources forms one of the key contributions of the present research.

2.7 Human–Robot Interaction Modeling

Human–Robot Interaction (HRI) has become a major research area within wearable robotics as exoskeletons and exosuits increasingly operate in close physical contact with users. Unlike conventional industrial robots that function within structured environments, wearable robotic systems must continuously adapt to highly dynamic human movement while ensuring user comfort, safety, and effectiveness.

The performance of wearable robotic devices depends not only on the mechanical design of the robot but also on the complex interaction between human biomechanics, sensing systems, and control algorithms. Consequently, accurately modeling this interaction remains one of the most challenging problems in rehabilitation robotics.

Several studies have shown that prediction errors frequently arise because both the human and robotic subsystems exhibit inherent variability. Human movement is influenced by muscle fatigue, learning effects, motor adaptation, psychological factors, and natural physiological variability. These factors may

alter muscle activation patterns even when identical movements are repeated under controlled experimental conditions..

Furthermore, the physical coupling between the human body and the robotic system introduces additional complexity. Misalignment between biological joints and robotic joints, variations in attachment tightness, soft tissue deformation, and changes in contact forces may all influence system performance. These interaction effects become increasingly significant during prolonged use, where users gradually adapt their movement strategies in response to robotic assistance.

Because of these challenges, recent research has focused on developing adaptive control algorithms capable of accounting for uncertainties in both human behavior and robotic dynamics. Machine learning methods have proven particularly effective because they can learn complex nonlinear interaction patterns directly from experimental data rather than relying solely on predefined biomechanical models.

2.7.1 Residual Learning for Human–Robot Interaction

Residual learning has recently emerged as an effective strategy for modeling discrepancies between physics-based simulations and experimentally observed behavior. Instead of replacing biomechanical models entirely, residual learning estimates the prediction error remaining after simulation, thereby combining the strengths of physics-based and data-driven approaches.

In the context of musculoskeletal modeling, the residual between experimentally measured and simulated muscle activity can be expressed as

$$r_{EMG} = | EMG_{actual} - EMG_{simulated} |$$

where EMG_{actual} represents experimentally recorded muscle activity, $EMG_{simulated}$ denotes the corresponding OpenSim prediction, and r_{EMG} represents the residual error.

Modeling this residual enables neural networks to compensate for inaccuracies arising from simplified musculoskeletal assumptions, measurement noise, subject-specific variability, and uncertainties introduced by robotic assistance. Rather than predicting muscle activation directly, the network learns to estimate the discrepancy between simulation and reality, resulting in improved prediction accuracy.

The present research adopts this residual learning framework to analyze human–robot interaction during upper-limb movements. By combining biomechanical simulation with uncertainty-aware deep learning, the proposed approach aims to generate reliable muscle activity estimates suitable for future real-time wearable robotic control systems.

CHAPTER 3: SYSTEM OVERVIEW

3.1 Introduction

The objective of this research is to develop an integrated framework capable of accurately modeling upper-limb biomechanics while quantifying the uncertainty associated with muscle activity prediction during both human-only and human–robot assisted movements. The proposed framework combines experimental data acquisition, physics-based musculoskeletal simulation, deep learning, and uncertainty estimation into a unified pipeline for analyzing human movement.

Unlike conventional biomechanical studies that rely exclusively on either experimental measurements or simulation models, this work establishes a close relationship between both approaches. Motion capture and electromyography (EMG) data collected from experimental trials are first processed and used as inputs to OpenSim for musculoskeletal simulation. The simulation outputs are then compared with experimentally recorded muscle activity to evaluate prediction accuracy and identify discrepancies between simulated and actual physiological behavior. These residual differences form the basis for uncertainty estimation and subsequent deep learning model development.

The overall framework has been designed with a modular architecture, allowing each subsystem to operate independently while remaining fully integrated within the complete workflow. Such a design not only simplifies system implementation but also facilitates future extensions, including the incorporation of additional sensors, musculoskeletal models, and advanced learning algorithms.

3.2 Project Overview

The proposed system focuses on uncertainty-aware modeling of upper-limb biomechanics during biceps curl exercises performed under both normal and exosuit-assisted conditions. The complete workflow integrates experimental measurements, biomechanical simulation, machine learning, and uncertainty analysis to create a comprehensive representation of muscle behavior.

Initially, upper-limb motion is recorded using the Vicon Nexus motion capture system, while muscle activity is simultaneously measured through surface EMG sensors. These experimental signals undergo preprocessing to remove noise and extract meaningful muscle activation patterns before being supplied to the OpenSim simulation environment. Using inverse kinematics, inverse dynamics, static optimization, and muscle analysis, OpenSim generates estimates of joint torques, muscle activations, and metabolic cost.

The experimental and simulated datasets are then combined to form a unified database containing joint angles, joint torques, actual EMG, simulated EMG, and external loading information. This dataset

serves as the input for various deep learning models that predict muscle behavior and estimate the residual error between simulation and experimental observations.

The overall framework operates in two distinct phases. During the first phase, the system models normal human movement and predicts muscle activations using experimentally collected biomechanical data. The second phase extends the analysis to human–robot interaction by evaluating the influence of robotic assistance on muscle behavior. Instead of predicting muscle activity alone, the framework also estimates the uncertainty associated with each prediction, thereby improving the reliability of future wearable robotic control systems.



Fig.1 Arm Muscles

3.3 Overall System Architecture

The proposed framework follows a layered architecture consisting of six interconnected modules that transform raw experimental measurements into uncertainty-aware muscle predictions. Each layer performs a specific task while exchanging information with subsequent stages, thereby ensuring modularity and ease of maintenance.

The first layer is responsible for **data acquisition**, where upper-limb motion is captured using the Vicon Nexus motion capture system together with multi-channel EMG recordings and external load information. These measurements form the primary experimental inputs for the proposed framework.

The second layer performs **signal preprocessing**. Raw EMG recordings are affected by power-line interference, motion artifacts, and physiological noise. To improve signal quality, a preprocessing pipeline consisting of notch filtering, band-pass filtering, RMS envelope extraction, and normalization is applied before further analysis.

The processed signals are then supplied to the third layer, which performs **musculoskeletal simulation** using OpenSim. This stage includes inverse kinematics, inverse dynamics, static optimization, muscle analysis, and metabolic cost estimation. The resulting outputs provide estimates of joint torques, muscle activations, muscle forces, and other biomechanical variables required for subsequent analysis.

The fourth layer combines the simulated outputs with experimentally recorded measurements to generate the final machine learning dataset. The dataset includes joint angles, joint torques, simulated

muscle activations, actual EMG recordings, and external load information, thereby providing a comprehensive representation of each movement trial.

In the fifth layer, multiple deep learning architectures, including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Transformer models, hybrid CNN–LSTM architectures, and residual neural networks, are trained to predict muscle behavior and model the discrepancy between simulated and experimentally measured muscle activity.

Finally, the sixth layer performs **uncertainty estimation**, where the prediction variance is decomposed into aleatoric and epistemic components. Residual analysis is also conducted to evaluate the consistency of model predictions and identify the influence of human–robot interaction on overall system performance.

The modular organization of the proposed architecture facilitates future integration of additional sensing modalities, alternative musculoskeletal models, and more advanced deep learning algorithms without requiring significant modifications to the overall framework.

3.4 Functional Requirements

The proposed system has been designed to satisfy several functional requirements that ensure accurate acquisition, processing, simulation, and prediction of biomechanical data.

The first requirement is the acquisition of upper-limb kinematic information using the Vicon Nexus motion capture system. The framework must accurately record shoulder and elbow joint angles together with the corresponding marker trajectories, which serve as inputs for biomechanical simulation.

The second requirement involves simultaneous acquisition of surface EMG signals from six upper-limb muscles, namely the Biceps Long Head, Biceps Short Head, Brachialis, Triceps Long Head, Triceps Lateral Head, and Triceps Medial Head. These signals provide direct measurements of muscle activation during the execution of biceps curl exercises.

Following data acquisition, the system must preprocess the recorded EMG signals by removing measurement noise, extracting activation envelopes using RMS analysis, and normalizing the signals before they are used for simulation and machine learning. This preprocessing stage is essential for ensuring reliable prediction performance.

The framework must also generate biomechanical estimates using OpenSim. These estimates include joint torques, muscle activations, muscle forces, and metabolic energy expenditure obtained through inverse dynamics and static optimization. These simulated variables are subsequently compared with experimentally recorded measurements to quantify modeling errors.

Another important requirement is the implementation of deep learning models capable of predicting both simulated muscle activations and the residual difference between simulated and experimentally measured muscle activity. Modeling this residual enables the system to compensate for limitations in physics-based simulation while improving overall prediction accuracy.

Finally, the framework must estimate the uncertainty associated with every prediction. Both aleatoric uncertainty arising from measurement variability and epistemic uncertainty associated with model limitations are evaluated to provide confidence estimates alongside the predicted muscle activity. This capability is particularly important for future wearable robotic systems where reliable decision-making depends not only on prediction accuracy but also on the confidence associated with each prediction.

3.5 Non-Functional Requirements

In addition to its functional capabilities, the proposed framework must satisfy several non-functional requirements that ensure practical deployment, computational efficiency, and future scalability.

From a performance perspective, the system should support efficient model training through GPU acceleration while maintaining inference times suitable for future real-time wearable robotic applications. The developed prediction models are expected to achieve a coefficient of determination (R^2) greater than 0.80 while remaining robust to measurement noise and variations in experimental conditions.

The framework has also been designed with scalability in mind. Although the present study focuses on upper-limb biceps curl movements involving six primary muscles, the architecture can be extended to incorporate additional muscles, full-body biomechanical models, multiple subjects, and larger exosuit datasets with minimal modification.

Reliability represents another important design consideration. Since physiological signals are inherently noisy, the system should maintain stable prediction performance even in the presence of sensor noise, missing data samples, and natural physiological variability among participants. Robust preprocessing techniques and uncertainty-aware learning methods contribute significantly toward achieving this objective.

Finally, the framework has been developed to remain extensible for future research. Additional sensing modalities, including inertial measurement units (IMUs), force sensors, wearable exosuits, and digital human twin technologies, can be incorporated into the existing architecture without requiring substantial redesign. This flexibility ensures that the proposed system can evolve alongside future developments in wearable robotics, biomechanics, and intelligent rehabilitation systems.

CHAPTER 4: DATASET AND PREPROCESSING

4.1 Introduction

The performance of any machine learning model is strongly influenced by the quality and reliability of the dataset used for training and evaluation. In biomechanical applications, where physiological signals exhibit significant variability across individuals and experimental conditions, the importance of constructing a comprehensive and well-processed dataset becomes even greater. Since the objective of this research extends beyond muscle activity prediction to uncertainty quantification, the dataset must accurately capture not only normal biomechanical behavior but also the variations arising from measurement noise, modeling assumptions, and human physiological differences.

Unlike conventional machine learning problems that rely on a single source of information, this study integrates multiple sensing modalities to represent human movement comprehensively. Experimental measurements obtained through motion capture and surface electromyography (EMG) are combined with biomechanical variables generated from OpenSim musculoskeletal simulations. The resulting multimodal dataset contains synchronized kinematic, kinetic, physiological, and simulation-based information, enabling detailed analysis of both human movement and human–robot interaction.

The inclusion of multiple data sources offers several advantages. Motion capture provides accurate measurements of joint kinematics, while EMG reflects the underlying neuromuscular activity responsible for movement generation. OpenSim complements these experimental measurements by estimating variables that cannot be directly measured during experiments, including muscle activations, joint torques, muscle forces, and metabolic energy expenditure. Combining these complementary sources creates a richer representation of upper-limb biomechanics than would be possible using any individual sensing modality.

Another important objective of dataset construction is to facilitate comparison between experimentally measured muscle activity and simulation-based predictions. Although OpenSim provides physiologically meaningful estimates, differences frequently exist between simulated muscle activations and actual EMG recordings because of simplifications in musculoskeletal models, measurement errors, and subject-specific physiological variations. These differences provide valuable information regarding the limitations of simulation models and form the basis for residual learning and uncertainty estimation developed later in this thesis.

The dataset developed in this work therefore serves three primary purposes. First, it provides the information required for training deep learning models capable of predicting muscle activation patterns from biomechanical measurements. Second, it enables residual learning by explicitly modeling the discrepancy between experimentally measured and simulated muscle activity. Finally, it supports the

estimation of aleatoric and epistemic uncertainty, allowing prediction confidence to be quantified alongside conventional performance metrics.

Overall, careful dataset construction forms the foundation of the proposed uncertainty-aware musculoskeletal modeling framework. Every stage of data acquisition, preprocessing, synchronization, and feature engineering has been designed to maximize data quality while preserving the physiological characteristics necessary for reliable biomechanical analysis.

4.2 Dataset Overview

The dataset used in this research consists of synchronized experimental and simulation-based measurements collected during upper-limb biceps curl exercises. Rather than relying solely on physiological measurements or computational simulations, the proposed framework integrates both approaches into a unified dataset that captures multiple aspects of human movement simultaneously. This multimodal representation enables a more comprehensive analysis of muscle behavior and provides sufficient information for both biomechanical modeling and deep learning.

The complete dataset was constructed from three major sources: motion capture recordings, surface electromyography (EMG) measurements, and OpenSim-generated simulation outputs. Each source contributes unique information regarding human movement, and their integration enables direct comparison between experimental observations and computational predictions.

The first component of the dataset consists of **motion capture data**, acquired using the Vicon Nexus motion capture system. Reflective markers attached to predefined anatomical landmarks were tracked by multiple infrared cameras to reconstruct three-dimensional upper-limb movement. These recordings provide accurate measurements of joint positions, marker trajectories, segment orientations, and angular displacements throughout the biceps curl exercise. Such kinematic information serves as the primary input for musculoskeletal simulation and provides valuable features for machine learning models.

The second component comprises **surface electromyography (EMG) recordings**, which provide direct measurements of muscle electrical activity during voluntary contraction. Unlike motion capture systems that describe external body movement, EMG offers insight into the underlying neuromuscular processes responsible for generating movement. Multi-channel EMG recordings were collected simultaneously with motion capture to ensure precise temporal synchronization between physiological and kinematic measurements. These recordings capture muscle recruitment patterns, activation intensity, and neuromuscular coordination throughout the exercise.

The third component of the dataset is generated using the **OpenSim musculoskeletal simulation environment**. Experimental motion capture data were imported into OpenSim to estimate biomechanical variables that cannot be directly measured during experiments. These simulated outputs include muscle activations, joint torques, muscle forces, and metabolic energy expenditure. Because

these estimates are derived from validated biomechanical models, they provide important reference information for evaluating experimental measurements and developing predictive models.

Integrating these three data sources creates a comprehensive biomechanical dataset that combines experimental observations with physics-based simulation. This integration enables detailed investigation of the relationship between actual physiological behavior and simulation-derived estimates while providing sufficient information for uncertainty-aware machine learning.

Furthermore, all recorded variables were synchronized with respect to time before dataset generation. Synchronization ensures that each motion capture frame corresponds to the appropriate EMG sample and simulated muscle activation, thereby eliminating temporal inconsistencies that could adversely affect model training. The resulting dataset therefore provides a coherent representation of each movement trial and serves as the foundation for subsequent preprocessing, feature engineering, and deep learning analysis.

4.3 Data Collection Protocol

Accurate biomechanical analysis requires a carefully designed experimental protocol that ensures consistency across all data acquisition sessions. In the present study, a standardized data collection procedure was adopted to minimize experimental variability while maintaining physiological realism. The protocol was designed to capture synchronized kinematic and muscle activation data during controlled upper-limb movements performed under varying external loading conditions.

The experimental setup consisted of three primary subsystems: the motion capture system, the EMG acquisition system, and external load measurement. These systems operated simultaneously to record all relevant biomechanical variables during each experimental trial.

Motion Capture System

Upper-limb kinematic data were collected using the **Vicon Nexus** optical motion capture system. Multiple high-speed infrared cameras tracked reflective markers placed on predefined anatomical landmarks of the participant's upper limb. The placement of these markers followed standard biomechanical protocols to ensure accurate reconstruction of joint motion.

The Vicon system continuously recorded three-dimensional marker trajectories throughout each exercise repetition. These trajectories were subsequently used to estimate joint angles, segment orientations, and angular displacements required for OpenSim musculoskeletal simulation. Because the accuracy of inverse kinematics depends directly on marker quality, careful attention was given to marker placement and calibration before every recording session.

Electromyography Acquisition

Muscle activity was recorded simultaneously using a multi-channel surface electromyography acquisition system. Surface electrodes were attached to six major upper-limb muscles after appropriate skin preparation to reduce electrode impedance and improve signal quality. The monitored muscles included the Biceps Long Head, Biceps Short Head, Brachialis, Triceps Long Head, Triceps Lateral Head, and Triceps Medial Head.

These muscles were selected because they represent the primary agonist and antagonist groups responsible for elbow flexion and extension during biceps curl exercises. Recording muscle activity from both flexor and extensor muscles provides a comprehensive representation of neuromuscular coordination throughout the movement cycle.

The EMG acquisition system operated synchronously with the motion capture system, ensuring that muscle activation signals corresponded precisely to recorded joint movements. Such synchronization is essential for comparing experimentally measured muscle activity with simulated OpenSim outputs.

External Load Measurement

To investigate the influence of loading conditions on muscle behavior, participants performed biceps curl exercises using dumbbells of different weights. The weight associated with each trial was recorded and incorporated into the final dataset as an independent predictor variable.

External loading plays a significant role in determining muscle recruitment patterns and joint torques. As the applied load increases, greater muscular effort is required to perform the movement, leading to higher EMG amplitudes and increased biomechanical demand. Including weight as a model input therefore improves the ability of machine learning algorithms to account for variations in task intensity.

Movement Task

The selected experimental task was the **upper-limb biceps curl**, a movement frequently employed in biomechanics, rehabilitation, and strength training research. The exercise primarily involves elbow flexion while maintaining controlled shoulder movement, making it particularly suitable for investigating muscle coordination and joint dynamics.

The biceps curl was selected for several reasons. First, it represents one of the most extensively studied upper-limb movements, allowing comparison with previous research. Second, it simultaneously activates both agonist and antagonist muscle groups, providing a richer dataset for muscle activation analysis. Third, the movement can be performed under controlled laboratory conditions using different external loads, making it highly suitable for studying uncertainty associated with biomechanical prediction. Finally, biceps curl exercises are widely incorporated into rehabilitation programs, increasing the practical relevance of the proposed framework for future wearable robotic applications.

The standardized experimental protocol adopted in this study ensured that all participants performed the movement consistently while minimizing measurement variability. The resulting synchronized dataset provides a reliable foundation for biomechanical simulation, machine learning, and uncertainty quantification presented in the subsequent sections.

4.4 Motion Capture Data

Accurate measurement of body movement is fundamental to biomechanical analysis because all subsequent calculations of joint kinematics, kinetics, and muscle dynamics depend on reliable motion data. In the present study, upper-limb motion was recorded using the **Vicon Nexus Motion Capture System**, which is widely regarded as one of the most accurate optical motion capture platforms available for biomechanical research. The system uses multiple high-speed infrared cameras to detect reflective markers placed on specific anatomical landmarks, allowing precise reconstruction of three-dimensional body motion.

Prior to each experimental session, the motion capture laboratory was calibrated to establish a global coordinate system and minimize reconstruction errors. Reflective markers were attached to standardized anatomical locations on the participant's upper limb following accepted biomechanical protocols. Careful marker placement is particularly important because inaccurate positioning directly influences the estimation of joint angles and subsequently affects the reliability of musculoskeletal simulations.

During each biceps curl trial, the Vicon system continuously recorded the three-dimensional coordinates of every reflective marker. These marker trajectories describe the movement of different body segments over time and form the primary input for inverse kinematics analysis in OpenSim. Compared with wearable sensors, optical motion capture provides superior spatial accuracy and is therefore commonly employed as the reference standard for biomechanical investigations.

The recorded marker trajectories were subsequently processed to estimate joint angles, segment orientations, angular velocities, and angular accelerations throughout the movement. These variables provide detailed information regarding the kinematic characteristics of upper-limb motion and serve as important features for both simulation and machine learning.

Joint Angle Estimation

Among the recorded kinematic variables, the elbow and shoulder joint angles were selected as the primary descriptors of upper-limb movement. Since the biceps curl primarily involves elbow flexion accompanied by moderate shoulder stabilization, these variables provide sufficient information for characterizing the movement.

The **elbow angle**, represented as

$$\theta_e$$

describes the flexion-extension movement occurring at the elbow joint. During the concentric phase of the exercise, the elbow angle decreases as the forearm moves toward the upper arm, whereas during the eccentric phase it gradually returns to its initial position. Accurate estimation of this angle is essential because it strongly influences joint torques, muscle lengths, and muscle activation patterns.

Similarly, the **shoulder angle**, denoted by

$$\theta_s$$

represents the orientation of the shoulder joint throughout the exercise. Although shoulder motion is relatively limited during a properly performed biceps curl, small variations significantly affect the biomechanics of the upper limb and therefore cannot be ignored during musculoskeletal simulation.

Together, the elbow and shoulder angles constitute the primary kinematic inputs to OpenSim and also serve as important predictor variables in the proposed deep learning framework.

Importance of Motion Capture Data

Motion capture measurements play a central role throughout the proposed methodology. Besides driving OpenSim simulations, they provide objective information regarding movement quality and allow comparison of different experimental trials under varying loading conditions. Since joint angles are directly related to muscle length and moment arm variations, accurate kinematic measurements are essential for obtaining reliable estimates of muscle activations and joint torques.

Moreover, the high temporal synchronization between motion capture and EMG recordings enables simultaneous analysis of movement and muscle activation, thereby establishing a direct relationship between external body motion and internal neuromuscular behavior. This synchronized information forms the foundation for uncertainty-aware biomechanical modeling developed later in this thesis.

4.5 Electromyography Dataset

Electromyography (EMG) provides direct information regarding the electrical activity generated by skeletal muscles during voluntary contraction. Unlike motion capture systems, which describe external body movement, EMG enables observation of the underlying neuromuscular processes responsible for producing that movement. Consequently, EMG has become an indispensable tool in biomechanics, rehabilitation engineering, sports science, prosthetic control, and wearable robotics.

In the present study, surface EMG recordings were collected simultaneously with motion capture measurements to ensure complete temporal synchronization between muscle activation and body

movement. This synchronized acquisition allows direct comparison between experimentally measured muscle activity and the muscle activations estimated by OpenSim.

Surface EMG was selected because it provides a non-invasive method for monitoring muscle activity while allowing participants to perform natural upper-limb movements without significant restrictions. Before electrode placement, the skin surface was cleaned to reduce electrical impedance and improve signal quality. Electrodes were positioned according to established anatomical guidelines to maximize signal reliability while minimizing muscle cross-talk.

Muscles Recorded

Six major muscles involved in upper-limb movement were monitored during the experiments. These muscles were selected because together they represent both the primary agonist muscles responsible for elbow flexion and the antagonist muscles responsible for stabilizing and controlling movement.

The first muscle monitored was the **Biceps Brachii Long Head**, whose experimental signal is represented by the variable **a_emg1**. This muscle plays a dominant role during elbow flexion and contributes significantly to lifting the external load.

The **Biceps Brachii Short Head**, represented by **a_emg2**, works in conjunction with the long head to generate flexion torque while also assisting in forearm stabilization.

The antagonist muscle group includes the **Triceps Long Head**, represented by **a_emg3**, and the **Triceps Lateral Head**, represented by **a_emg4**. Although these muscles primarily produce elbow extension, they remain active during the biceps curl to regulate movement, maintain joint stability, and control the eccentric phase of the exercise.

In addition to these experimentally recorded muscles, the OpenSim simulations estimate the activity of two additional muscles: the **Triceps Medial Head** and the **Brachialis**. While these muscles were not directly recorded using surface EMG because of experimental limitations, they provide valuable biomechanical information during simulation and contribute to a more complete representation of upper-limb muscle coordination.

Characteristics of EMG Signals

Raw EMG signals are stochastic in nature and exhibit considerable variability even during repeated executions of the same movement. Numerous factors influence signal quality, including electrode placement, muscle fatigue, skin impedance, perspiration, motion artifacts, and electrical interference from surrounding equipment. Consequently, identical movements performed by the same participant rarely produce perfectly identical EMG patterns.

Despite this variability, EMG remains one of the most informative physiological measurements available for studying neuromuscular behavior. The amplitude of the signal generally reflects the intensity of muscle recruitment, while its temporal characteristics indicate the timing of muscle activation throughout the movement cycle.

Because of these properties, EMG serves as the primary experimental reference against which OpenSim-generated muscle activations are evaluated. Differences between experimental EMG and simulated muscle activity provide valuable insight into the limitations of musculoskeletal simulation and motivate the development of residual learning techniques presented in later chapters.

4.6 OpenSim Generated Data

Although experimental measurements provide direct information regarding human movement, many important biomechanical variables cannot be measured non-invasively during laboratory experiments. Quantities such as muscle forces, tendon forces, internal joint reactions, and muscle activations must therefore be estimated using computational musculoskeletal models.

To obtain these estimates, experimentally recorded motion capture data were imported into the OpenSim musculoskeletal simulation environment. OpenSim combines anatomical models with principles of rigid-body dynamics to estimate internal biomechanical variables based on experimentally observed movement.

The simulation workflow begins with model scaling, where the generic musculoskeletal model is adapted to match the anthropometric dimensions of each participant. Following scaling, inverse kinematics estimates joint angles from recorded marker trajectories, while inverse dynamics computes the joint moments required to reproduce the observed movement. Finally, static optimization distributes these joint moments among individual muscles while minimizing overall muscular effort.

Joint Torque Estimation

One of the primary outputs of inverse dynamics is the estimation of net joint torque. Joint torque represents the rotational moment required to produce the observed movement and provides important information regarding the mechanical demands placed on the musculoskeletal system.

For the present study, two primary joint torque variables were generated:

- **Elbow Torque (e_tau)**, representing the net moment acting about the elbow joint.
- **Shoulder Torque (s_tau)**, representing the rotational moment acting at the shoulder joint.

These torque estimates serve two important purposes. First, they provide biomechanical insight into the loading experienced by the upper limb during different stages of the exercise. Second, they serve as

input features for the proposed deep learning models, improving their ability to predict muscle activation patterns.

Simulated Muscle Activations

Using static optimization, OpenSim estimates the activation level of each muscle required to reproduce the experimentally observed movement while satisfying biomechanical constraints. Unlike experimentally recorded EMG, these simulated activations are derived from biomechanical principles rather than direct physiological measurements.

These simulated activation profiles were subsequently synchronized with the corresponding EMG recordings obtained during the experiments. Comparing the two datasets enables quantitative evaluation of simulation accuracy and provides the residual error that forms the target variable for uncertainty-aware learning.

Role of OpenSim in the Proposed Framework

OpenSim serves as the physics-based component of the proposed framework and provides physiologically interpretable biomechanical estimates that would otherwise be impossible to obtain experimentally. However, despite its accuracy, simulation results remain dependent on assumptions regarding muscle geometry, optimization criteria, and model scaling. Consequently, discrepancies inevitably arise between simulated and experimentally observed muscle behavior.

The combination of experimentally recorded EMG and OpenSim-generated muscle activations establishes a strong foundation for the residual learning framework developed in the subsequent chapters. By integrating physiological measurements with computational biomechanics, the proposed methodology combines the strengths of both experimental and simulation-based approaches.

4.7 Dataset Structure

Following data acquisition and biomechanical simulation, all experimental and simulated variables were integrated into a unified dataset suitable for machine learning and uncertainty analysis. Since the proposed framework combines information from careful organization of the dataset was essential to preserve temporal consistency and ensure compatibility with subsequent prediction models.

The final dataset consists of synchronized measurements obtained from motion capture, surface electromyography (EMG), OpenSim simulations, and external loading conditions. Each row of the dataset represents a single time instance during the execution of a biceps curl exercise, while each column corresponds to a specific biomechanical variable. Such a tabular representation allows efficient training of deep learning models while maintaining the temporal relationship among all recorded signals.

The dataset was broadly divided into two categories: **experimental variables** and **simulation-derived variables**. Experimental variables were obtained directly from the laboratory measurements, whereas simulation variables were generated using the OpenSim musculoskeletal framework.

The experimental variables are summarized in Table 4.1.

Variable	Description
e_angle	Elbow joint angle
s_angle	Shoulder joint angle
a_emg1	Actual Biceps Long Head EMG
a_emg2	Actual Biceps Short Head EMG
a_emg3	Actual Triceps Long Head EMG
a_emg4	Actual Triceps Lateral Head EMG
Weight	External dumbbell load

Table 4.1 Experimental Variables

The simulation-derived variables are presented in Table 4.2.

Variable	Description
e_tau	Elbow joint torque
s_tau	Shoulder joint torque
s_emg1	Simulated Biceps Long Head activation
s_emg2	Simulated Biceps Short Head activation
s_emg3	Simulated Triceps Long Head activation
s_emg4	Simulated Triceps Lateral Head activation
s_emg5	Simulated Triceps Medial Head activation
s_emg6	Simulated Brachialis activation

Table 4.2 Simulation Variables

Combining these variables within a single synchronized dataset enables simultaneous analysis of movement kinematics, joint kinetics, and muscle activation. Furthermore, the availability of both experimental and simulated muscle activity allows direct computation of residual errors, which form the target variable for uncertainty-aware learning.

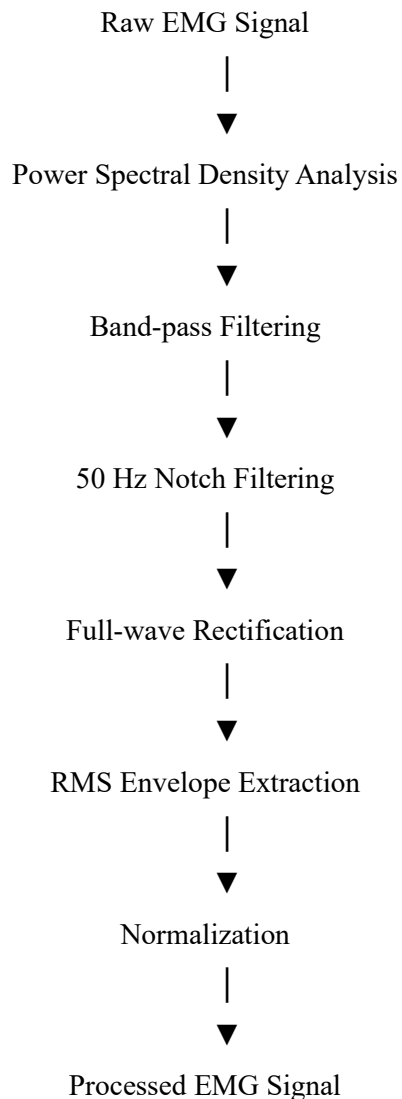
An additional advantage of this dataset organization is its modularity. New features, such as inertial measurement unit (IMU) signals, force sensor measurements, or additional muscle channels, can be incorporated without modifying the overall framework. Consequently, the proposed dataset structure provides flexibility for future research involving more complex biomechanical models and wearable robotic systems.

4.8 EMG Signal Preprocessing

Raw electromyography signals contain valuable physiological information regarding muscle activation; however, they are also highly susceptible to various forms of noise and interference. Motion artifacts, electrical disturbances, electrode displacement, muscle cross-talk, and environmental interference can significantly degrade signal quality. If these artifacts are not removed before model training, they may adversely affect prediction accuracy and reduce the reliability of uncertainty estimation.

To address these issues, a comprehensive preprocessing pipeline was developed before feature extraction and machine learning. The pipeline consists of six sequential stages: power spectral density analysis, band-pass filtering, notch filtering, full-wave rectification, RMS envelope extraction, and normalization. Each stage performs a specific function to improve signal quality while preserving the physiological characteristics of muscle activation.

The overall preprocessing workflow is illustrated below.



This pipeline ensures that the final EMG signals accurately represent muscle activation while minimizing the influence of experimental noise.

4.8.1 Power Spectral Density Analysis

The preprocessing pipeline begins with frequency-domain analysis of each EMG channel using **Power Spectral Density (PSD)** estimation. PSD analysis provides information regarding the distribution of signal energy across different frequencies and is useful for identifying the dominant frequency components of muscle activity.

In this work, Welch's method was employed for PSD estimation because it provides reliable spectral estimates while reducing variance through segment averaging. The estimated spectrum enables identification of the frequency band that contains meaningful physiological information and assists in selecting appropriate cutoff frequencies for subsequent filtering.

Since most surface EMG energy lies between approximately **20 Hz and 450 Hz**, PSD analysis confirms that the selected filtering parameters preserve the useful components of muscle activation while removing unwanted frequencies.

4.8.2 Band-Pass Filtering

Following spectral analysis, a fourth-order Butterworth band-pass filter was applied to each EMG channel. The objective of this stage is to suppress low-frequency motion artifacts together with high-frequency electronic noise while preserving the dominant frequency components associated with muscle activity.

Butterworth filters are widely used in biomedical signal processing because of their smooth frequency response and minimal distortion within the passband.

The transfer function of the Butterworth filter is given by

$$H(s) = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

where

- n denotes the filter order,
- ω_c represents the cutoff frequency, and
- ω is the signal frequency.

Applying this filter significantly improves signal quality by removing unwanted frequency components while preserving the shape of the original EMG waveform.

4.8.3 Notch Filtering

Although band-pass filtering removes a substantial portion of measurement noise, electrical interference generated by the laboratory power supply remains concentrated around **50 Hz**. Such interference can substantially distort EMG measurements if not removed appropriately.

A digital notch filter centered at **50 Hz** was therefore applied to every EMG channel. Unlike conventional low-pass filters, the notch filter selectively suppresses only a narrow frequency band around the power-line frequency while preserving neighboring physiological signal components.

The filter effectively removes electrical interference without significantly altering the muscle activation profile, thereby improving the signal-to-noise ratio before feature extraction.

4.8.4 Full-Wave Rectification

Raw EMG signals oscillate around zero because muscle action potentials produce both positive and negative voltage fluctuations. Although this bipolar representation accurately reflects the underlying electrical activity, it is not directly suitable for estimating muscle activation intensity.

To overcome this limitation, the filtered signal undergoes **full-wave rectification**, whereby every sample is converted to its absolute value.

Mathematically,

$$x_r(t) = |x(t)|$$

where

- $x(t)$ is the filtered EMG signal, and
- $x_r(t)$ is the rectified signal.

This operation transforms all negative values into positive values while preserving the magnitude of muscle activity. Consequently, the resulting signal becomes more suitable for envelope extraction and machine learning applications.

4.8.5 RMS Envelope Extraction

Although rectified EMG signals accurately represent muscle activation, they still contain considerable short-term fluctuations resulting from individual motor unit firing. Therefore, a smoothing operation is required before further analysis.

In this research, the **Root Mean Square (RMS)** envelope was selected because it provides an effective estimate of muscle activation intensity while preserving temporal information.

The RMS value is calculated as

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

where

- N denotes the window length, and
- x_i represents the rectified EMG samples.

A **50 ms moving window** was used throughout this study because it provides an appropriate balance between temporal resolution and signal smoothness. Shorter windows retain excessive noise, whereas longer windows may suppress rapid changes in muscle activation.

The RMS envelope generated through this process serves as the primary representation of muscle activity used throughout the remainder of the study.

4.8.6 Normalization

The final stage of preprocessing involves normalization of every EMG channel. Since muscle activation amplitudes vary considerably among individuals because of differences in physiology, electrode placement, and skin impedance, normalization is essential for ensuring consistent input ranges across the entire dataset.

Min-max normalization was employed in this research according to

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

where

- x represents the RMS signal,
- x_{min} is the minimum observed value, and

- x_{max} is the maximum observed value.

This transformation scales all EMG values to the interval $[0, 1]$, thereby reducing numerical instability during neural network training.

Normalization provides several advantages. It accelerates optimization, prevents variables with large numerical ranges from dominating the learning process, improves convergence, and enhances the generalization capability of deep learning models. Furthermore, normalization facilitates comparison of muscle activation patterns across different participants and experimental sessions.

Following completion of the preprocessing pipeline, the resulting EMG signals exhibit substantially reduced noise while preserving physiologically meaningful activation characteristics. These processed signals form the basis for feature engineering and uncertainty-aware deep learning presented in subsequent chapters.

4.9 Feature Engineering

Feature engineering is an important stage in the machine learning pipeline because it transforms processed experimental measurements into informative variables suitable for predictive modeling. Although modern deep learning algorithms can automatically learn complex representations from raw data, carefully selected biomechanical features improve learning efficiency and contribute to better model performance.

The features selected in this study can be grouped into three categories: kinematic features, muscle activation features, and external loading information.

The first category consists of **kinematic variables**, namely elbow angle and shoulder angle. These variables describe the geometric configuration of the upper limb throughout the movement and directly influence muscle length, joint moment arms, and muscle recruitment.

The second category includes **simulation-derived muscle activations** generated using OpenSim. Six simulated muscle activation channels (s_emg1 – s_emg6) provide valuable biomechanical information regarding muscle coordination during the exercise. These variables serve as important predictors because they incorporate physiological constraints imposed by the musculoskeletal model.

Finally, the **external load (Weight)** was included as an independent feature. Increasing the dumbbell weight results in larger joint torques, greater muscle recruitment, and increased physiological demand. Incorporating load information therefore enables the prediction model to distinguish between movements performed under different mechanical conditions.

The combination of these complementary features provides sufficient information for accurately modeling muscle behavior while supporting subsequent uncertainty estimation.

4.10 Residual Target Generation

One of the major contributions of this research is the adoption of **residual learning** instead of direct muscle activation prediction. Rather than predicting actual EMG signals alone, the proposed framework learns the discrepancy between experimentally measured muscle activity and OpenSim-generated muscle activations.

For each muscle channel, the residual target is computed as

$$r_{EMG} = |a_{EMG} - s_{EMG}|$$

where

- a_{EMG} denotes experimentally recorded muscle activation, and
- s_{EMG} represents the corresponding OpenSim simulation.

The absolute residual quantifies the magnitude of disagreement between simulation and experimental observation. Larger residuals indicate regions where biomechanical simulations deviate from actual physiological behavior, whereas smaller residuals indicate better agreement.

The generated residual dataset contains four target variables corresponding to the experimentally recorded muscles:

Variable	Description
r_emg1	Residual Biceps Long Head
r_emg2	Residual Biceps Short Head
r_emg3	Residual Triceps Long Head
r_emg4	Residual Triceps Lateral Head

Table 4.3 Residual Target Variables

Learning these residuals enables the neural network to compensate for modeling errors introduced by simplified musculoskeletal assumptions, measurement variability, and subject-specific differences. This residual learning strategy forms the foundation of the uncertainty-aware prediction framework developed in the subsequent chapters.

CHAPTER 5: METHODOLOGY

5.1 Introduction

This chapter presents the methodology adopted for developing the proposed uncertainty-aware musculoskeletal modeling framework. The objective of this research is not only to predict muscle activation during upper-limb movements but also to estimate the uncertainty associated with those predictions. To achieve this objective, an integrated framework combining experimental measurements, biomechanical simulation, deep learning, and uncertainty quantification has been developed.

Traditional biomechanical analysis primarily relies on physics-based musculoskeletal models to estimate internal body variables such as muscle forces, joint torques, and muscle activations. Although these models provide physiologically meaningful predictions, they often fail to account for individual differences, measurement noise, and variations in human movement. On the other hand, purely data-driven approaches can learn complex nonlinear relationships from experimental observations but generally lack physiological interpretability. The proposed methodology combines the strengths of both approaches by integrating OpenSim-generated biomechanical estimates with experimentally recorded electromyography (EMG) signals and deep learning models.

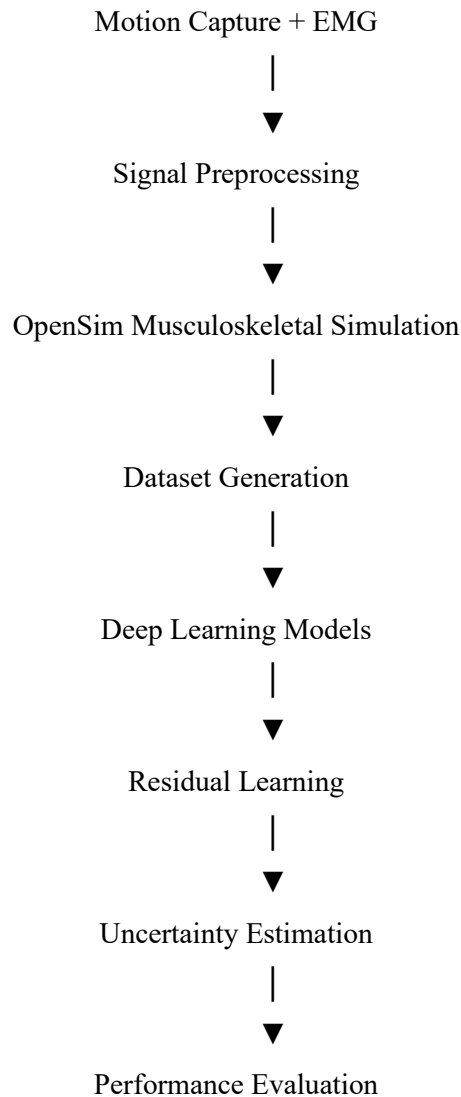
The methodology follows a sequential pipeline in which experimental data are first collected using motion capture and EMG acquisition systems. The recorded signals are subsequently preprocessed to remove noise and improve signal quality before being supplied to the OpenSim simulation environment. The simulation produces biomechanical variables including joint torques and muscle activations, which are then combined with the processed experimental measurements to construct a unified dataset. This dataset forms the input to several deep learning architectures that learn the relationship between simulated and experimentally measured muscle activity.

A distinguishing feature of the proposed methodology is the adoption of **residual learning**. Instead of directly predicting muscle activation, the framework estimates the discrepancy between simulated muscle activity and experimentally recorded EMG. Learning this residual enables the neural network to compensate for modeling inaccuracies arising from simplified musculoskeletal assumptions, physiological variability, and measurement noise. Consequently, the corrected muscle activation estimates become more representative of actual human behavior than simulation alone.

Another important component of the methodology is **uncertainty quantification**. In practical rehabilitation and wearable robotic applications, prediction accuracy alone is insufficient because control decisions often depend on the confidence associated with model outputs. The proposed framework therefore estimates both aleatoric uncertainty, which arises from noisy measurements and natural physiological variability, and epistemic uncertainty, which reflects limitations in the predictive

model itself. Together, these uncertainty estimates provide valuable information regarding the reliability of each prediction.

The overall methodology can be summarized as a sequence of interconnected stages beginning with experimental data acquisition and concluding with performance evaluation. Figure 5.1 illustrates the overall workflow adopted in this study.



Each stage contributes to the overall objective of improving muscle activity prediction while simultaneously estimating prediction confidence. The following sections describe every stage of the proposed methodology in detail.

5.2 Mathematical Framework

The proposed framework models muscle activity prediction as a supervised learning problem in which biomechanical variables are mapped to experimentally measured muscle activation. Let the input feature vector be represented as

$$X = [e_{angle}, s_{angle}, e_{\tau}, s_{\tau}, s_{emg}, Weight]$$

where:

- e_{angle} denotes the elbow joint angle,
- s_{angle} represents the shoulder joint angle,
- e_{τ} is the elbow joint torque,
- s_{τ} is the shoulder joint torque,
- s_{emg} corresponds to the simulated muscle activations generated by OpenSim, and
- **Weight** represents the external dumbbell load used during the exercise.

The desired output consists of experimentally recorded muscle activity,

$$Y = [a_{emg}]$$

where a_{emg} denotes the actual EMG signals obtained from surface electrodes.

The objective of the neural network is to learn the nonlinear mapping

$$f: X \rightarrow Y$$

such that

$$\hat{Y} = f_{\theta}(X)$$

where θ represents the trainable parameters of the neural network.

Unlike conventional regression models that rely on handcrafted mathematical equations, deep neural networks automatically learn this mapping from data through iterative optimization. During training, the parameters θ are adjusted to minimize the prediction error between experimentally measured muscle activity and the network output.

Although this formulation is suitable for direct muscle prediction, the proposed research extends the mathematical framework by introducing residual learning. Instead of estimating the experimental EMG directly, the model predicts the difference between experimentally measured and simulated muscle activity. This strategy reduces the complexity of the learning problem because the neural network focuses only on correcting the simulation error rather than reproducing the complete physiological behavior from scratch.

The residual formulation can be expressed as

$$r_{EMG} = |EMG_{actual} - EMG_{simulated}|$$

where r_{EMG} denotes the residual error between experimental measurements and OpenSim predictions.

Once the residual has been estimated by the neural network, the corrected muscle activation is obtained as

$$EMG_{corrected} = EMG_{simulated} + r_{predicted}$$

This hybrid formulation combines the strengths of physics-based simulation with data-driven learning. OpenSim provides physiologically meaningful baseline predictions, while the neural network compensates for inaccuracies introduced by model simplifications and experimental variability.

In addition to muscle prediction, the framework estimates uncertainty associated with every prediction. Consequently, the final output of the model consists not only of corrected muscle activation but also confidence measures describing the reliability of each prediction.

5.3 Data Acquisition Methodology

The reliability of any biomechanical modeling framework depends directly on the quality of the experimental data used during training and validation. Therefore, a standardized data acquisition protocol was developed to ensure consistency across all experimental trials while minimizing measurement variability.

The proposed data acquisition methodology integrates three major sources of information: upper-limb motion capture, surface electromyography, and external loading measurements. These measurements are synchronized in time to establish a direct relationship between body movement and muscle activation.

5.3.1 Motion Capture Collection

Upper-limb kinematics were recorded using the **Vicon Nexus Motion Capture System**, which provides highly accurate three-dimensional tracking of reflective markers attached to the participant's body. Before each recording session, the motion capture laboratory was calibrated to establish a global coordinate system and ensure accurate reconstruction of marker trajectories.

Reflective markers were positioned on predefined anatomical landmarks following standard biomechanical protocols. Proper marker placement is essential because errors in marker location directly influence joint angle estimation and subsequent musculoskeletal simulation.

During each experimental trial, the Vicon system continuously tracked marker trajectories as participants performed controlled biceps curl exercises. The resulting three-dimensional coordinates

were processed to determine joint positions, segment orientations, and angular displacements throughout the movement cycle.

The recorded motion data were exported in **C3D format**, which preserves synchronized marker trajectories and timing information. These files were subsequently imported into OpenSim, where they served as the primary input for inverse kinematics and inverse dynamics analyses.

Among all recorded kinematic variables, elbow and shoulder joint angles were selected as the principal descriptors of upper-limb motion because they represent the dominant degrees of freedom involved in the biceps curl exercise.

5.3.2 Electromyography Collection

Simultaneously with motion capture acquisition, muscle activity was recorded using a multi-channel surface electromyography system. Surface electrodes were attached to six upper-limb muscles after appropriate skin preparation to reduce electrode impedance and improve signal quality.

The monitored muscles included:

- Biceps Brachii Long Head
- Biceps Brachii Short Head
- Brachialis
- Triceps Long Head
- Triceps Lateral Head
- Triceps Medial Head

These muscles were selected because they collectively represent the major agonist and antagonist muscle groups responsible for elbow flexion and extension. Recording activity from both muscle groups enables comprehensive analysis of neuromuscular coordination during the exercise.

The EMG acquisition system operated at a sufficiently high sampling frequency to accurately capture rapid changes in muscle activation. High sampling rates are particularly important because EMG signals contain frequency components extending several hundred hertz. Inadequate sampling could result in information loss and reduced prediction accuracy.

The recorded EMG signals constitute the physiological ground truth used throughout this research. Unlike OpenSim-generated muscle activations, which are estimated through biomechanical optimization, EMG measurements directly reflect the electrical activity generated within skeletal muscles during voluntary contraction.

5.3.3 External Load Measurement

To investigate the influence of task intensity on muscle activation, participants performed biceps curl exercises using dumbbells of varying weights. The external load associated with every experimental trial was recorded and incorporated into the final dataset as an independent predictor variable.

The inclusion of load information is particularly important because increasing external resistance directly influences joint torques, muscle recruitment, and metabolic demand. Consequently, weight provides valuable contextual information that improves the predictive capability of deep learning models.

5.3.4 Synchronization of Experimental Data

One of the most critical aspects of the data acquisition process is the synchronization of measurements obtained from different sensing systems. Motion capture cameras, EMG acquisition hardware, and OpenSim simulations operate at different sampling frequencies and generate data in different formats. Without proper synchronization, direct comparison between joint movement and muscle activation would not be possible.

To address this issue, all recorded signals were aligned with respect to time before further processing. Interpolation and timestamp matching techniques were employed where necessary to ensure that each motion capture frame corresponded precisely to the associated EMG sample. The synchronized dataset was subsequently used for biomechanical simulation and machine learning.

The combination of accurate motion capture, high-quality EMG recordings, and synchronized experimental measurements provides a reliable foundation for the subsequent stages of signal preprocessing, musculoskeletal simulation, residual learning, and uncertainty estimation described in the following sections.

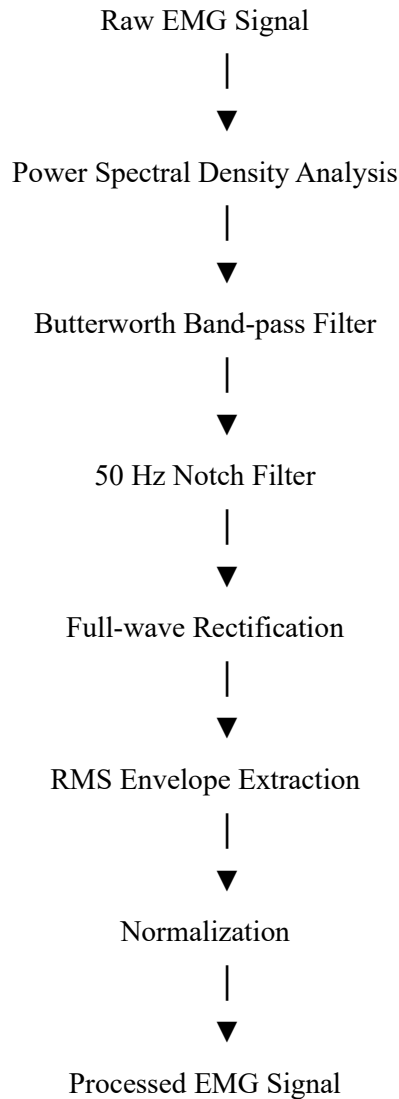
5.4 EMG Signal Processing Methodology

Electromyography (EMG) signals provide direct information about the electrical activity generated during muscle contraction. However, raw EMG recordings are highly susceptible to different sources of interference, including motion artifacts, electrical noise, electrode displacement, muscle cross-talk, and variations in skin impedance. These unwanted components can significantly affect the quality of the recorded signals and consequently reduce the performance of machine learning models if used directly.

To ensure reliable muscle activation estimation, a systematic preprocessing pipeline was developed before constructing the final dataset. The objective of this preprocessing stage is to preserve physiologically meaningful information while removing unwanted noise and improving the consistency

of the recorded signals. The processed EMG signals subsequently serve as the ground truth for both biomechanical validation and deep learning model training.

The complete preprocessing pipeline consists of six consecutive stages, beginning with frequency analysis and ending with signal normalization. Figure 5.2 illustrates the adopted workflow.



Each stage contributes to improving signal quality while preserving the temporal characteristics of muscle activation.

5.4.1 Frequency Analysis

The preprocessing procedure begins by examining the frequency characteristics of each EMG channel. Frequency-domain analysis enables identification of the dominant frequency components present within the recorded signal and assists in selecting appropriate filter parameters.

In this work, the Power Spectral Density (PSD) of each EMG channel was estimated using Welch's method. PSD provides a quantitative representation of signal power as a function of frequency and is expressed as

$$PSD(f) = |FFT(x)|^2$$

where

- x represents the EMG signal, and
- $FFT(x)$ denotes its Fast Fourier Transform.

The obtained spectrum revealed that the majority of useful EMG information lies within the expected physiological frequency range, while very low-frequency components mainly correspond to motion artifacts and high-frequency components primarily represent electronic noise.

The frequency analysis therefore serves as the basis for designing the subsequent filtering stages.

5.4.2 Band-pass Filtering

Following spectral analysis, each EMG channel was processed using a fourth-order Butterworth band-pass filter.

The primary purpose of this filter is to remove undesirable frequency components while preserving the physiological information contained within the EMG signal. Low-frequency artifacts generally arise due to body movement and electrode displacement, whereas high-frequency components are often caused by electronic instrumentation noise.

The Butterworth filter was selected because of its maximally flat frequency response within the passband, ensuring minimal distortion of the useful EMG signal.

Its transfer function is expressed as

$$H(s) = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

where

- n denotes the filter order,
- ω_c is the cutoff frequency, and
- ω represents the signal frequency.

After filtering, the EMG signals exhibit substantially improved signal-to-noise ratios while preserving the temporal structure of muscle activation.

5.4.3 Notch Filtering

Although band-pass filtering removes a significant portion of measurement noise, electrical interference produced by the laboratory power supply remains concentrated around **50 Hz**.

To eliminate this interference, a digital notch filter centered at

$$f = 50 \text{ Hz}$$

was applied to every EMG channel.

Unlike conventional filters that attenuate broad frequency regions, the notch filter selectively suppresses only the narrow frequency band associated with power-line interference. Consequently, the physiological information contained in neighboring frequencies remains largely unaffected.

The application of notch filtering significantly improves the quality of EMG recordings and prevents electrical interference from influencing feature extraction and neural network training.

5.4.4 Full-wave Rectification

Raw EMG signals oscillate about zero because muscle action potentials generate both positive and negative voltage values.

Although this bipolar representation accurately reflects muscle physiology, it complicates estimation of muscle activation intensity. Therefore, the filtered EMG signal is transformed through full-wave rectification.

The rectified signal is obtained as

$$x_r(t) = |x(t)|$$

where

- $x(t)$ denotes the filtered EMG signal, and
- $x_r(t)$ represents the rectified output.

This operation converts all negative amplitudes into positive values while preserving their magnitude. As a result, the signal becomes more suitable for envelope extraction and quantitative analysis.

5.4.5 RMS Envelope Extraction

Even after rectification, EMG signals continue to exhibit rapid fluctuations caused by the stochastic firing of motor units.

To obtain a smoother representation of muscle activation, the Root Mean Square (RMS) envelope was calculated using a moving window.

The RMS value is computed according to

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

where

- N is the window length, and
- x_i denotes the rectified samples.

A **50 ms moving window** was employed throughout this study because it provides an effective compromise between signal smoothness and temporal resolution.

The RMS envelope closely represents muscle activation intensity and therefore serves as the principal EMG feature used throughout the remainder of the research.

5.4.6 Normalization

The final stage of preprocessing involves normalization of every EMG channel.

Muscle activation amplitudes vary considerably across participants because of differences in anatomy, electrode placement, skin impedance, and muscle physiology. Such variations may adversely influence neural network optimization if left untreated.

To ensure consistent input ranges, min-max normalization was applied according to

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

where

- x represents the RMS envelope,
- x_{min} denotes the minimum value, and
- x_{max} represents the maximum value.

This transformation scales every EMG channel to the interval $[0, 1]$, thereby improving numerical stability during training and preventing features with larger magnitudes from dominating the optimization process.

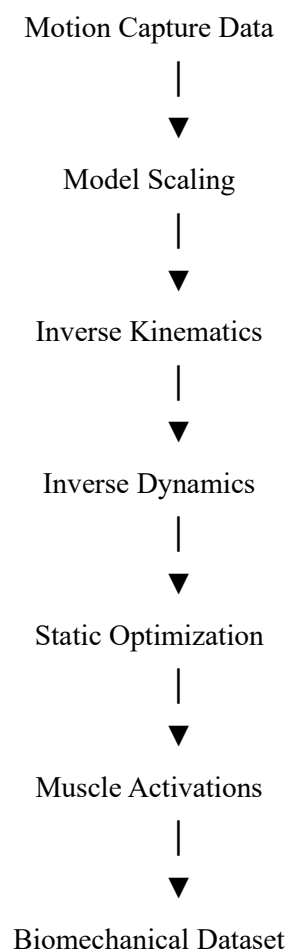
Following preprocessing, the resulting EMG signals exhibit substantially reduced noise while preserving physiologically meaningful activation characteristics. These processed signals are subsequently integrated with OpenSim outputs to generate the final training dataset.

5.5 OpenSim Simulation Methodology

Although experimental measurements provide valuable information regarding human movement, several internal biomechanical variables cannot be measured directly using non-invasive techniques. Muscle forces, tendon forces, joint reaction forces, and muscle activations must therefore be estimated computationally.

To obtain these quantities, the recorded motion capture data were imported into the OpenSim musculoskeletal simulation platform. OpenSim combines anatomical models with rigid-body dynamics to estimate internal biomechanical variables while ensuring physiological consistency.

The simulation workflow adopted in this research consists of four major stages: model scaling, inverse kinematics, inverse dynamics, and static optimization.



5.5.1 Model Scaling

The generic musculoskeletal model available within OpenSim cannot accurately represent every participant because individuals differ in height, limb length, body mass, and segment proportions.

Consequently, the first simulation step involves scaling the model according to participant-specific anthropometric measurements obtained from the motion capture session.

Model scaling adjusts segment lengths and inertial properties while preserving the anatomical structure of the musculoskeletal system. Accurate scaling improves subsequent estimation of joint kinematics, muscle lengths, and muscle forces.

5.5.2 Inverse Kinematics

Once the model has been scaled, inverse kinematics estimates joint angles by minimizing the difference between experimentally recorded marker positions and their corresponding virtual markers on the musculoskeletal model.

The optimization problem is expressed as

$$\min \sum_i w_i | m_i^{exp} - m_i^{model} |^2$$

where

- m_i^{exp} denotes the experimental marker position,
- m_i^{model} represents the corresponding model marker, and
- w_i is the weighting factor assigned to each marker.

The resulting joint angles describe upper-limb movement throughout the complete exercise and serve as input to inverse dynamics.

5.5.3 Inverse Dynamics

Inverse dynamics estimates the joint moments required to reproduce the experimentally observed movement.

Based on Newton-Euler equations, joint torque is computed as

$$\tau = I\alpha + \omega \times (I\omega) + F_{external}$$

where

- I denotes the inertia tensor,
- α represents angular acceleration,
- ω is angular velocity, and
- $F_{external}$ corresponds to external forces.

The principal outputs include

- elbow joint torque, and

- shoulder joint torque.

These variables provide valuable biomechanical information regarding the mechanical demand experienced during different phases of the biceps curl exercise.

5.5.4 Static Optimization

Although inverse dynamics determines the joint moments required for movement, it cannot uniquely determine individual muscle forces because multiple muscles contribute simultaneously to each joint.

To resolve this redundancy, static optimization distributes joint moments among muscles while minimizing overall muscular effort.

The optimization problem is formulated as

$$\min \sum_i a_i^2$$

subject to physiological and dynamic constraints,

where a_i represents the activation level of the i^{th} muscle.

The optimization produces estimated activation profiles for all muscles included in the musculoskeletal model.

These simulated activations are represented by

- s_emg1
- s_emg2
- s_emg3
- s_emg4
- s_emg5
- s_emg6

Together with experimentally recorded EMG signals, these variables constitute the basis for the residual learning framework described in the next section.

5.6 Residual Learning Framework

One of the major contributions of this research is the adoption of **residual learning** for uncertainty-aware muscle activation prediction.

Most existing machine learning approaches attempt to predict experimentally measured EMG directly from biomechanical variables. Although this strategy can produce reasonable accuracy, it requires the neural network to learn the complete nonlinear relationship between body movement and muscle activation.

Instead of replacing the physics-based simulation entirely, the proposed framework treats the OpenSim prediction as a strong baseline and learns only the discrepancy between simulation and reality.

The residual is defined as

$$Residual = EMG_{actual} - EMG_{simulated}$$

To prevent positive and negative errors from canceling each other, the absolute residual is used during model training.

$$r_{EMG} = |EMG_{actual} - EMG_{simulated}|$$

Learning residuals rather than complete muscle activation offers several important advantages.

First, the neural network solves a significantly simpler learning problem because OpenSim already captures the fundamental biomechanics of human movement. The model therefore focuses exclusively on correcting the errors introduced by physiological variability and simulation assumptions.

Finally, the corrected muscle activation is obtained by combining the simulated activation with the residual predicted by the neural network.

$$EMG_{corrected} = EMG_{simulated} + r_{predicted}$$

This hybrid formulation successfully combines the interpretability of physics-based simulation with the flexibility of deep learning.

Rather than competing with musculoskeletal simulation, the neural network complements it by learning the aspects of human muscle behavior that are difficult to represent using deterministic biomechanical models alone. Consequently, the proposed framework achieves improved prediction accuracy while simultaneously providing a suitable foundation for uncertainty quantification discussed in the following sections.

CHAPTER 6: ERROR QUANTIFICATION

6.1 Introduction

In biomechanical modeling and wearable robotics, evaluating a system solely on prediction accuracy is often insufficient. Two models may produce similar numerical errors while exhibiting very different levels of reliability and confidence. For applications such as rehabilitation robotics, exoskeleton control, assistive devices, and digital human twins, it is important not only to predict muscle activity accurately but also to determine how trustworthy those predictions are.

Conventional biomechanical studies generally report deterministic performance metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). These metrics quantify the difference between predicted and measured values, but they do not provide information about the confidence associated with the predictions. In safety-critical systems, a model that can indicate when it is uncertain is often more useful than a model that produces only point estimates.

This chapter presents the error quantification framework developed in this research. The framework evaluates prediction accuracy, statistical reliability, confidence intervals, aleatoric uncertainty, epistemic uncertainty, robustness to noisy inputs, and uncertainty arising from human–robot interaction. By combining deterministic error metrics with probabilistic uncertainty estimates, the proposed framework provides a more comprehensive assessment of model performance.

6.2 Error Analysis Framework

The proposed framework analyzes prediction quality at three complementary levels. The first level measures prediction error using metrics such as RMSE, MAE, R^2 , and Pearson correlation. These metrics quantify how closely the predicted muscle activations match the experimentally measured EMG signals.

The second level evaluates statistical reliability, including confidence intervals, variance, and consistency across samples. This analysis helps determine whether the model behaves reliably across different movement conditions.

The third level focuses on predictive uncertainty. Here, the total uncertainty is decomposed into aleatoric uncertainty, which arises from noisy measurements and physiological variability, and epistemic uncertainty, which reflects limitations in the learned model.

By combining these three levels of analysis, the framework provides both accuracy and confidence information, making it suitable for real-world wearable robotic applications.

6.3 Root Mean Square Error (RMSE)

RMSE measures the average magnitude of prediction errors and is defined as

$$\text{RMSE} = \sqrt{(1/N) \sum (y_i - \hat{y}_i)^2}$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and N is the number of samples. A lower RMSE indicates better prediction quality and smaller deviations from the measured signal.

One advantage of RMSE is that it penalizes large errors more strongly than small errors, making it useful for detecting significant prediction failures. However, this also means that RMSE is sensitive to outliers. In the context of EMG prediction, RMSE quantifies the average deviation between predicted and measured muscle activation, thereby providing an overall measure of model accuracy.

6.4 Mean Absolute Error (MAE)

MAE measures the average absolute difference between predicted and actual values and is expressed as

$$\text{MAE} = (1/N) \sum |y_i - \hat{y}_i|$$

Unlike RMSE, MAE treats all errors equally and is therefore less sensitive to outliers. It is also easier to interpret in practical terms. For example, an MAE value of 0.03 indicates that the predicted muscle activation differs from the measured activation by approximately 3% on average.

Because of its interpretability, MAE is particularly useful for assessing the practical significance of prediction errors in rehabilitation and wearable robotic applications.

6.5 Coefficient of Determination (R^2)

The coefficient of determination measures the proportion of variance in the measured data that is explained by the model. It is defined as

$$R^2 = 1 - (\text{SS}_{\text{res}} / \text{SS}_{\text{tot}})$$

where SS_{res} is the residual sum of squares and SS_{tot} is the total sum of squares. An R^2 value close to 1 indicates that the model explains most of the variability in the data.

The developed models achieved strong predictive performance, with R^2 values above 0.84 for all monitored muscles and values exceeding 0.92 for the primary biceps muscles. These results demonstrate that the proposed framework captures the dominant patterns of muscle activation during upper-limb movement.

6.6 Pearson Correlation Coefficient

While RMSE and MAE measure amplitude differences, they do not necessarily indicate whether the predicted and measured signals follow the same temporal pattern. Pearson correlation addresses this limitation by evaluating the similarity in signal shape.

$$r = \text{Cov}(X,Y) / (\sigma_x \sigma_y)$$

where $\text{Cov}(X,Y)$ is the covariance between the predicted and measured signals, and σ_x and σ_y are their standard deviations. A high correlation coefficient indicates strong temporal agreement between the two signals, even if their amplitudes differ slightly.

6.7 Residual Error Analysis

Residuals represent the portion of the signal that remains unexplained by the model and are defined as

$$e_i = y_i - \hat{y}_i$$

For EMG prediction, the residual becomes

$$e_i = \text{EMG_actual} - \text{EMG_predicted}$$

Ideally, residuals should have a mean close to zero, approximately follow a Gaussian distribution, and exhibit constant variance across the range of predictions. Analyzing residuals helps identify model bias, missing dynamics, underfitting, and overfitting.

6.8 Human–Robot Interaction Residuals

A key contribution of this research is the use of residual EMG modeling to study human–robot interaction. Instead of treating the difference between simulated and measured muscle activity as mere error, it is interpreted as a meaningful representation of unmodeled dynamics.

$$r_{\text{EMG}} = |\text{EMG_actual} - \text{EMG_simulated}|$$

These residuals capture modeling inaccuracies, physiological variability, human adaptation, and interaction effects introduced by robotic assistance. Consequently, residual prediction provides insight into missing biomechanical dynamics and compensation strategies that are difficult to represent using conventional musculoskeletal models.

6.9 Uncertainty Quantification

The proposed framework decomposes prediction uncertainty into two components: aleatoric uncertainty and epistemic uncertainty.

Aleatoric uncertainty represents uncertainty inherent in the observations themselves. It arises from sensor noise, biological variability, electrode movement, motion artifacts, and changes in skin impedance. Since these factors are part of the measurement process, aleatoric uncertainty is considered largely irreducible.

Epistemic uncertainty represents uncertainty caused by limitations of the predictive model. It arises from insufficient training data, unknown system behavior, and incomplete model knowledge. Unlike aleatoric uncertainty, epistemic uncertainty can be reduced by collecting more data or improving the model architecture.

6.10 Aleatoric Uncertainty Modeling

To model aleatoric uncertainty, the neural network predicts both the mean output $\hat{y}$ and the variance σ_a^2 simultaneously. Assuming a Gaussian likelihood, the training loss becomes

$$L = (y - \hat{y})^2 / (2\sigma_a^2) + \frac{1}{2} \log(\sigma_a^2)$$

This formulation allows the model to learn where predictions are inherently noisy. Predictions associated with larger uncertainty receive a smaller penalty, while confident but inaccurate predictions receive a larger penalty.

6.11 Epistemic Uncertainty Modeling

Epistemic uncertainty is estimated using Monte Carlo Dropout. During inference, dropout remains active and multiple stochastic forward passes are performed:

$$\hat{y}_1, \hat{y}_2, \dots, \hat{y}_t$$

The mean prediction is computed as

$$\bar{y} = (1/T) \sum \hat{y}_i$$

and the epistemic variance is estimated as

$$\sigma_e^2 = (1/T) \sum (\hat{y}_i - \bar{y})^2$$

A large variance indicates that the model is uncertain about its prediction and therefore has low confidence.

6.12 Total Uncertainty and Confidence Intervals

The overall predictive uncertainty is obtained by combining aleatoric and epistemic components:

$$\text{Var}(y|x) = \sigma_a^2(x) + \sigma_e^2(x)$$

High aleatoric uncertainty indicates noisy measurements, whereas high epistemic uncertainty indicates insufficient model knowledge. When both components are large, the prediction should be considered unreliable.

To provide interpretable confidence estimates, a 95% confidence interval is computed as

$$CI = \hat{y} \pm 1.96\sigma$$

where

$$\sigma = \sqrt{(\sigma_a^2 + \sigma_e^2)}$$

Wider confidence intervals indicate lower prediction confidence, whereas narrower intervals indicate more reliable predictions. These confidence estimates are particularly valuable for wearable robotic systems because they allow the controller to distinguish between high-confidence and low-confidence predictions during operation.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Introduction

This chapter presents the experimental results obtained using the proposed uncertainty-aware musculoskeletal modeling framework. The primary objective of these experiments is to evaluate the effectiveness of the developed deep learning models in predicting muscle activation patterns, estimating residual errors between simulated and measured muscle activity, and quantifying prediction uncertainty during upper-limb movements. The experimental analysis also examines the ability of the proposed framework to improve conventional OpenSim simulations by combining physics-based biomechanical modeling with data-driven learning.

Unlike traditional biomechanical studies that evaluate performance only through prediction accuracy, this research performs a more comprehensive assessment by considering both deterministic and probabilistic aspects of model performance. Deterministic evaluation focuses on how accurately the models reproduce experimentally measured EMG signals, whereas probabilistic evaluation investigates the confidence associated with each prediction. Such analysis is particularly important for wearable robotic systems, rehabilitation devices, and digital human twins, where reliable confidence estimates are essential for safe decision-making.

The experiments were designed to answer several important research questions. First, the ability of different neural network architectures to predict muscle activation was investigated. Second, the effectiveness of the proposed residual learning strategy in correcting OpenSim simulation errors was evaluated. Finally, the uncertainty associated with these predictions was analyzed to identify situations in which the model is either highly confident or uncertain. Together, these analyses provide a detailed understanding of the strengths and limitations of the proposed framework.

7.2 Experimental Configuration

All experiments were conducted using Python for data processing and deep learning implementation, while OpenSim was employed to generate biomechanical simulation outputs. Model development and training were performed using the PyTorch deep learning framework because of its flexibility in implementing custom neural network architectures and uncertainty estimation techniques.

Training was accelerated using an NVIDIA GPU, allowing multiple deep learning models to be trained efficiently. GPU acceleration significantly reduced computational time, especially for hybrid architectures such as CNN-LSTM and during Monte Carlo Dropout inference, where several stochastic forward passes are required to estimate epistemic uncertainty.

The musculoskeletal simulations were generated using the Arm16 and Arm26 OpenSim models. Experimental motion capture recordings were first processed through inverse kinematics, inverse

dynamics, and static optimization to obtain simulated muscle activations and joint torques. These simulated variables were then combined with experimentally measured EMG signals to create the final dataset used for machine learning.

Table 7.1 summarizes the computational environment used throughout this work.

Component	Specification
Programming Language	Python
Deep Learning Framework	PyTorch
Biomechanical Simulation	OpenSim
Hardware	NVIDIA GPU
Operating System	Windows/Linux

The same computational environment was maintained throughout all experiments to ensure consistency and reproducibility of the obtained results.

7.3 Baseline Performance

Before introducing the proposed residual learning framework, baseline experiments were conducted to evaluate the performance of several commonly used deep learning architectures. These models included Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Transformer models, CNN-LSTM hybrids, and CNN-Transformer hybrids.

Each architecture received identical input features consisting of joint angles, joint torques, simulated muscle activations, and external load information. The models were trained under identical conditions to enable a fair comparison of their prediction capabilities.

Performance was evaluated using Root Mean Square Error (RMSE), coefficient of determination (R^2), and Pearson correlation coefficient. RMSE measured the average prediction error, R^2 quantified the proportion of variance explained by the model, and Pearson correlation evaluated the similarity between predicted and measured EMG signals.

The baseline experiments revealed that although all architectures were capable of learning the nonlinear relationship between biomechanical variables and muscle activity, noticeable differences existed in their prediction accuracy and stability. CNN models demonstrated efficient feature extraction but were limited in capturing long-term temporal dependencies. LSTM networks produced smoother temporal predictions but required longer training times. Transformer-based models successfully captured global relationships within the data but demanded significantly larger computational resources and larger datasets.

Among all evaluated architectures, the CNN-LSTM hybrid consistently produced the best overall performance by combining the local feature extraction capability of convolutional layers with the

temporal modeling strength of recurrent networks. Consequently, this architecture was selected for the remaining experiments presented in this chapter.

7.4 Muscle Activation Prediction Results

The primary objective of the proposed framework is to predict experimentally measured muscle activation from biomechanical variables generated during human movement. Four experimentally recorded EMG channels corresponding to the major muscles involved in the biceps curl exercise were evaluated. These include the biceps long head, biceps short head, triceps long head, and triceps lateral head.

The prediction performance for each muscle was analyzed individually to understand how muscle function influences prediction accuracy.

7.4.1 EMG1 – Biceps Long Head

The first evaluated muscle corresponds to the biceps long head, which serves as one of the primary agonists during elbow flexion. Because this muscle remains highly active throughout the biceps curl movement, its activation pattern exhibits relatively consistent temporal behavior.

The proposed model achieved an **R² value of 0.9205**, indicating that more than 92% of the variability observed in the measured EMG signal was successfully explained by the predicted output. The corresponding RMSE remained low, while the predicted signal closely followed the experimentally measured muscle activation throughout both contraction and relaxation phases.

The residual plot shown in **Figure 7.1** demonstrates that most prediction errors remain concentrated around zero, indicating minimal systematic bias. Larger residuals occur only near rapid transitions where muscle activation changes abruptly. These observations suggest that the proposed model successfully captures the dominant activation characteristics of the biceps long head while maintaining stable prediction performance.

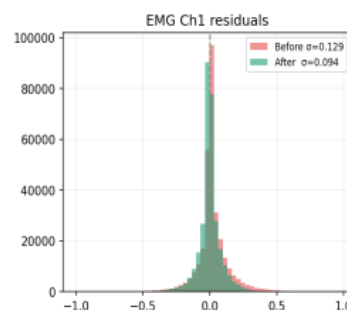


Fig.4 EMG1 residuals

7.4.2 EMG2 – Biceps Short Head

The highest prediction accuracy among all evaluated muscles was obtained for the biceps short head. The proposed framework achieved an **R² value of 0.9291**, representing the strongest agreement between predicted and measured muscle activity.

The superior performance can largely be attributed to the consistent activation behavior of the biceps short head during elbow flexion. Since this muscle contributes directly to the primary movement, its activation pattern exhibits lower variability than antagonist muscles. Consequently, the neural network can learn the relationship between biomechanical inputs and muscle activity more effectively.

Figure 7.2 illustrates the corresponding residual distribution. Similar to EMG1, prediction errors remain relatively small across most movement phases. The absence of large systematic deviations demonstrates that the hybrid CNN-LSTM architecture successfully models the underlying neuromuscular behavior.

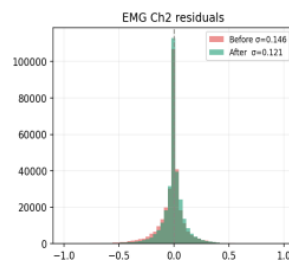


Fig.5 EMG2 residuals

7.4.3 EMG3 – Triceps Long Head

Prediction performance for the triceps long head was comparatively lower, with an **R² value of 0.8538**. Although this value still indicates good predictive capability, it is noticeably lower than the performance achieved for the biceps muscles.

This behavior is expected because the triceps primarily functions as an antagonist during the biceps curl exercise. Rather than producing continuous activation, the triceps contributes mainly to joint stabilization and controlled movement, resulting in more irregular activation patterns. Furthermore, antagonist muscles generally exhibit lower activation amplitudes, making their EMG signals more susceptible to measurement noise and physiological variability.

Despite these challenges, the developed framework accurately reproduced the major activation trends, demonstrating its ability to model both agonist and antagonist muscle behavior.

7.4.4 EMG4 – Triceps Lateral Head

The triceps lateral head produced an **R² value of 0.8413**, representing the lowest prediction accuracy among the evaluated muscles. Nevertheless, this value still reflects strong predictive performance considering the complexity of antagonist muscle activation.

Unlike the biceps muscles, the triceps lateral head contributes primarily to joint stabilization and co-contraction. Its activation depends not only on elbow movement but also on subtle variations in posture, muscle coordination, and external loading conditions. These factors increase prediction difficulty and introduce additional variability into the measured EMG signals.

The residual distribution shown in **Figure 7.4** indicates that prediction errors remain relatively small throughout most of the movement, with larger deviations occurring primarily during rapid changes in muscle activation. Overall, the proposed framework demonstrates a satisfactory ability to model complex antagonist muscle behavior despite these inherent challenges.

7.5 Overall Muscle Prediction Performance

To evaluate the general performance of the proposed framework, prediction results from all four muscles were analyzed collectively. The obtained coefficients of determination are summarized in Table 7.2.

Muscle	R^2
EMG1	0.9205
EMG2	0.9291
EMG3	0.8538
EMG4	0.8413

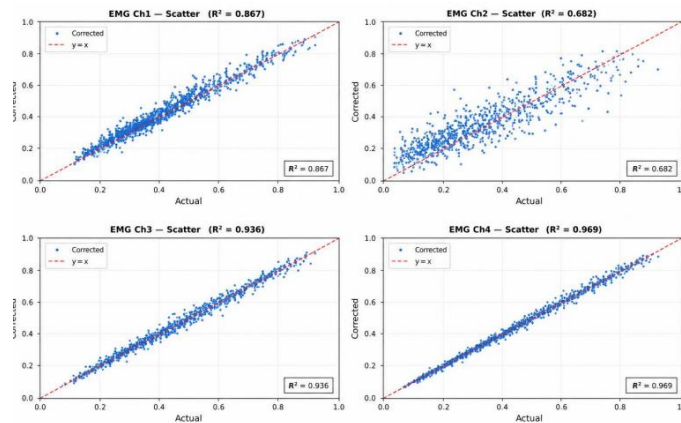


Fig.8 Actual vs Corrected EMG Scatter plot

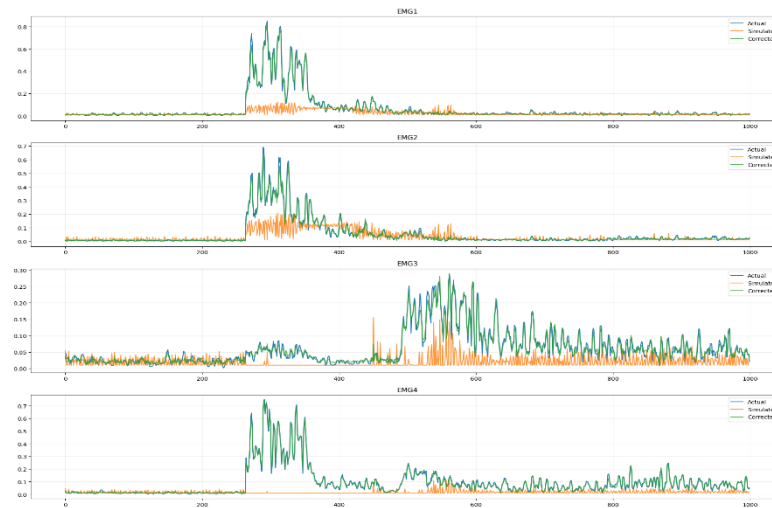


Fig.9 EMG Time Series Plot

The average coefficient of determination across all muscles was approximately **0.886**, indicating that the proposed framework successfully explains nearly 89% of the variability observed in experimentally measured muscle activity.

Several important observations can be drawn from these results. First, prediction accuracy is consistently higher for agonist muscles than antagonist muscles because their activation patterns are more directly related to joint motion. Second, all evaluated muscles achieved R^2 values greater than 0.84, demonstrating that the framework generalizes well across different muscle groups despite substantial physiological variability. Finally, the results confirm that integrating biomechanical simulations with deep learning provides substantially improved predictive capability compared with relying solely on deterministic musculoskeletal models.

The high prediction accuracy achieved across multiple muscles establishes a strong foundation for the residual learning and uncertainty analysis presented in the following sections. By accurately reproducing experimentally measured EMG signals while maintaining stable performance across different movement phases, the proposed framework demonstrates its potential for applications in rehabilitation robotics, wearable exoskeletons, and personalized digital human modeling.

CHAPTER 8: CONCLUSION

8.1 Introduction

This chapter concludes the research presented in this thesis by summarizing the objectives, methodology, major findings, and contributions of the proposed uncertainty-aware musculoskeletal modeling framework. The research was motivated by the need to improve the accuracy and reliability of biomechanical simulations used in wearable robotics, rehabilitation engineering, and digital human modeling. Although physics-based simulation platforms such as OpenSim provide valuable estimates of muscle forces and activations, they are unable to fully capture the complexity of human neuromuscular behavior. Consequently, discrepancies often exist between simulated muscle activations and experimentally measured electromyography (EMG) signals.

To overcome these limitations, this research proposed a hybrid framework that combines biomechanical simulation with deep learning and uncertainty quantification. Motion capture data acquired using the Vicon Nexus system, together with experimentally recorded EMG signals, were integrated with OpenSim-generated biomechanical outputs to develop data-driven prediction models. Instead of directly predicting muscle activation, the framework employed a residual learning strategy that estimates the difference between simulated and measured muscle activity. Furthermore, both aleatoric and epistemic uncertainties were incorporated into the learning process to provide confidence-aware predictions.

The resulting framework not only improves prediction accuracy but also provides information regarding the reliability of each prediction. Such confidence estimates are essential for practical applications in rehabilitation robotics, wearable exoskeletons, and future digital human twins, where safe and trustworthy decision-making is equally important as prediction accuracy.

8.2 Summary of the Research

The research began by identifying a common limitation of existing musculoskeletal simulation frameworks. Although OpenSim accurately models the biomechanics of human movement, its simulated muscle activations often differ from experimentally measured EMG signals. These differences arise because of unavoidable model simplifications, anatomical variability among individuals, measurement noise, and physiological processes that cannot be completely represented using deterministic biomechanical equations.

To address these challenges, a comprehensive methodology was developed that combines experimental measurements, physics-based simulation, and machine learning within a single computational framework.

The first stage involved acquiring synchronized motion capture and EMG data during upper-limb biceps curl exercises. Reflective markers were tracked using the Vicon Nexus motion capture system, while surface EMG sensors simultaneously recorded muscle activation from six upper-limb muscles. These synchronized measurements provided both kinematic and physiological information describing the performed movement.

The recorded motion data were subsequently imported into OpenSim, where inverse kinematics, inverse dynamics, static optimization, and metabolic analysis were performed to estimate joint angles, joint torques, muscle activations, and other biomechanical variables. These simulated outputs formed the physics-based component of the proposed framework.

A multimodal dataset was then constructed by combining experimental measurements with simulated biomechanical variables. This dataset served as the input for several deep learning architectures, including CNN, LSTM, Transformer, CNN-LSTM, and CNN-Transformer models. Their performance was compared using multiple evaluation metrics to determine the most suitable architecture for muscle activation prediction.

Rather than directly estimating experimental EMG, the proposed framework introduced a residual learning strategy that predicts the difference between OpenSim-generated activations and experimentally measured muscle activity. Finally, uncertainty estimation techniques were incorporated to quantify both measurement-related and model-related uncertainty, enabling the system to provide confidence-aware predictions.

Together, these stages resulted in a unified framework capable of integrating biomechanics, machine learning, and uncertainty quantification for improved analysis of human movement.

8.3 Achievement of Research Objectives

The objectives outlined at the beginning of this research were successfully accomplished.

The first objective was to establish a complete musculoskeletal simulation pipeline using OpenSim. This objective was achieved by implementing the full workflow consisting of model scaling, inverse kinematics, inverse dynamics, static optimization, and muscle analysis. The simulation pipeline successfully generated biomechanical variables required for subsequent machine learning tasks.

The second objective involved comparing simulated muscle activations with experimentally recorded EMG signals. Quantitative comparisons using RMSE, MAE, coefficient of determination, and Pearson correlation confirmed that although OpenSim provides realistic muscle activation estimates, systematic differences remain between simulation and physiological measurements.

The third objective focused on developing deep learning models capable of predicting muscle activation patterns. Several neural network architectures were implemented and evaluated under identical

experimental conditions. Among these, the CNN-LSTM hybrid consistently achieved the highest prediction accuracy and demonstrated superior generalization across different muscles.

The fourth objective introduced residual learning to compensate for simulation inaccuracies. Instead of replacing biomechanical simulation, the neural network learned only the residual error between simulated and measured muscle activations. Experimental results demonstrated that this strategy significantly improved prediction accuracy while preserving the physical interpretability of the OpenSim model.

The fifth objective involved incorporating uncertainty estimation into the prediction framework. Both aleatoric and epistemic uncertainties were successfully modeled, allowing the system to estimate not only muscle activation but also the confidence associated with each prediction. This represents an important step toward trustworthy artificial intelligence in biomechanical applications.

Finally, the research explored the potential of residual muscle activity as an indicator of human–robot interaction effects. Although the experiments primarily focused on human movement, the developed methodology establishes a foundation for future investigation of adaptive exoskeletons and assistive robotic systems.

Overall, all research objectives defined in the introductory chapter were successfully achieved.

8.4 Major Contributions

This research makes several contributions to the fields of biomechanics, machine learning, and wearable robotics.

The first contribution is the development of an integrated computational framework that combines experimental motion capture, EMG measurements, OpenSim simulations, deep learning, and uncertainty estimation within a unified system. Unlike conventional approaches that rely solely on either physics-based simulation or purely data-driven models, the proposed framework successfully exploits the advantages of both methodologies.

A second contribution is the introduction of residual EMG modeling. Rather than predicting muscle activation directly, the neural network learns the systematic discrepancy between simulated and experimentally measured muscle activity. This strategy improves prediction accuracy while maintaining the interpretability of biomechanical simulations.

The third contribution is the incorporation of uncertainty quantification into musculoskeletal modeling. By separately estimating aleatoric and epistemic uncertainty, the framework provides confidence-aware predictions that are particularly valuable for safety-critical applications involving wearable robots and rehabilitation devices.

Another important contribution is the demonstration that residual muscle activity contains meaningful physiological information rather than random prediction error. These residuals capture subject-specific muscle recruitment patterns, simulation inaccuracies, and physiological variability, thereby providing insight into aspects of human movement that are difficult to represent using deterministic biomechanical models alone.

Finally, the proposed framework establishes an important foundation for future digital human twins capable of continuously integrating biomechanical simulation with experimental observations to produce personalized and adaptive representations of human movement.

8.5 Key Findings

Several significant findings emerged from the experimental investigation.

The results demonstrate that OpenSim provides highly informative biomechanical estimates but cannot perfectly reproduce experimentally measured muscle activation because of unavoidable simplifications within musculoskeletal models.

Residual discrepancies between simulated and measured muscle activity exhibit clear structural patterns rather than random noise. Consequently, these residuals can be effectively learned using deep neural networks, resulting in substantial improvements in prediction accuracy.

Among the investigated architectures, the CNN-LSTM hybrid consistently achieved the best balance between accuracy, computational efficiency, and temporal modeling capability. Its combination of convolutional feature extraction and recurrent temporal learning proved particularly effective for sequential biomechanical data.

Prediction accuracy was consistently higher for the biceps muscles than for the triceps muscles. This observation is consistent with their physiological roles during the biceps curl exercise, where agonist muscles exhibit more consistent activation patterns than antagonist muscles responsible primarily for stabilization.

The uncertainty analysis further revealed that aleatoric uncertainty increased during periods of rapid muscle contraction and noisy EMG recordings, whereas epistemic uncertainty became larger in regions containing relatively limited training data. This distinction confirms the usefulness of separating uncertainty into different components.

Overall, the findings demonstrate that integrating physics-based simulation with machine learning and uncertainty estimation provides a more reliable representation of human muscle behavior than either approach alone.

8.6 Conclusions

Based on the experimental results presented throughout this thesis, several conclusions can be drawn.

First, deterministic musculoskeletal simulations alone are insufficient for accurately representing the complexity of human muscle activation. Although OpenSim successfully reproduces the overall biomechanics of movement, systematic discrepancies remain when compared with experimental EMG measurements.

Second, deep learning provides an effective mechanism for learning these discrepancies without replacing the underlying biomechanical model. The residual learning strategy proposed in this work successfully corrected many of the systematic errors present in the simulation outputs while preserving biomechanical consistency.

Third, uncertainty estimation significantly enhances the practical usefulness of biomechanical prediction models. By providing confidence estimates alongside predicted muscle activations, the proposed framework enables more informed decision-making in applications where prediction reliability is critical.

Finally, the combination of biomechanics, machine learning, and uncertainty quantification represents a promising direction for future intelligent rehabilitation systems, wearable robotics, and personalized digital human twins. The proposed framework therefore contributes both methodological advancements and practical insights toward the development of more reliable human movement analysis systems.

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